

WIN
SERVER

AZ-800

Administering Windows Server Hybrid Core Infrastructure

Interview Questions & Answers

Complete Windows Server Administration Preparation Guide

2025 EDITION

300

Questions

10

Topics

Beginner

to Advanced

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Microsoft Windows Server Certification Preparation Guide

AZ-800 Certification Preparation | 2025 Edition

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PREFACE & INTRODUCTION

Welcome to the AZ-800: Administering Windows Server Hybrid Core Infrastructure Interview Questions & Answers guide — your comprehensive companion for Microsoft Windows Server certification preparation.

The AZ-800 exam (which also covers the legacy 70-740 exam topics) validates skills in Windows Server administration including Active Directory, DNS, DHCP, Hyper-V, storage, networking, security, high availability, and hybrid cloud integration with Azure.

This guide contains 300 carefully crafted interview questions and answers spanning all major exam domains — from Windows Server fundamentals to advanced hybrid cloud scenarios. Whether preparing for the Microsoft certification exam or a Windows Server Administrator technical interview, this guide provides comprehensive coverage.

Topics Covered: Windows Server Core Concepts • Active Directory Domain Services • DNS & DHCP • Hyper-V & Virtualization • Storage & File Services • Network Policy & Remote Access • Windows Server Security • High Availability & Clustering • Monitoring & Troubleshooting • Hybrid & Cloud Integration

How to use this guide: Read through questions progressively. Try to answer each question before reading the provided answer. Focus on understanding the WHY behind configurations, not just the commands.

Best of luck in your AZ-800 certification journey!

— Akhilesh Chaudhari

Section 1: Windows Server Core Concepts

Questions 1 – 30

Q1 What is Windows Server 2022 and what are its key editions?

Answer : Windows Server 2022 is Microsoft's latest server operating system offering enhanced security, hybrid cloud capabilities, and performance improvements. Key editions: Datacenter (unlimited VMs, all features, software-defined networking), Standard (2 VMs per license, core features), and Essentials (for small businesses, up to 25 users). Datacenter includes exclusive features like Storage Spaces Direct, Software-defined networking, and Shielded VMs.

Q2 What are the installation options for Windows Server 2022?

Answer : Windows Server 2022 offers two installation options: Server Core (no GUI – command-line and PowerShell only, smaller footprint, lower attack surface, fewer updates) and Server with Desktop Experience (full GUI). Microsoft recommends Server Core for most roles. A third option, Nano Server, is available only as a container base image. You can switch from Desktop Experience to Core but not back without reinstallation.

Q3 What is Windows Server Core and what are its advantages?

Answer : Server Core is a minimal Windows Server installation without the graphical shell (Windows Explorer, Control Panel, etc.). Advantages: reduced attack surface (fewer services and components), lower disk and memory footprint, fewer reboots for updates, and better suited for automation. Administration is done via PowerShell, Windows Admin Center, or Server Manager remotely. Most server roles are supported on Server Core.

Q4 What is Windows Admin Center (WAC)?

Answer : Windows Admin Center is a free, browser-based management tool for Windows Servers, clusters, and hyper-converged infrastructure. It runs locally or as a gateway service. Features: server management (roles, services, events, performance), Azure hybrid services integration (Azure Backup, Site Recovery, Monitor), cluster management, and extension support. WAC replaces many Server Manager and MMC functions with a modern web interface.

Q5 What is the Server Manager in Windows Server?

Answer : Server Manager is the built-in management console in Windows Server with Desktop Experience. It provides a centralized dashboard for: adding/removing roles and features (via Add Roles and Features Wizard), managing local and remote servers, viewing events, services, best practices analyzer results, and performance data. Server Manager can manage multiple servers simultaneously from one console.

Q6 What are Windows Server Roles and Features?

Answer : Roles are major software components providing specific server functions — e.g., DNS Server, DHCP Server, Active Directory Domain Services, File and Storage Services, Web Server (IIS), Hyper-V. Features are supporting software components — e.g., .NET Framework, BitLocker, Failover Clustering, Remote Server Administration Tools (RSAT). Roles and Features are installed via Server Manager or PowerShell `Install-WindowsFeature` cmdlet.

Q7 What is PowerShell and why is it important for Windows Server?

Answer : PowerShell is Microsoft's cross-platform command-line shell and scripting language built on .NET. It's the primary automation tool for Windows Server administration. Key concepts: cmdlets (verb-noun format: `Get-Service`, `Set-Item`), pipeline (pass objects between cmdlets), modules (collections of cmdlets), remoting (manage remote servers), desired state configuration (DSC), and scheduled jobs. All Windows Server roles have corresponding PowerShell modules.

Q8 What is PowerShell Remoting?

Answer : PowerShell Remoting allows running PowerShell commands on remote computers. Built on WinRM (Windows Remote Management), which uses WS-Management protocol (HTTP port 5985, HTTPS port 5986). Enable with: `Enable-PSRemoting`. Connect with: `Enter-PSSession` (interactive), `Invoke-Command` (run commands on multiple computers). Supports CredSSP for double-hop authentication. Requires WinRM service running and firewall rules allowing port 5985/5986.

Q9 What is Windows Remote Management (WinRM)?

Answer : WinRM is Microsoft's implementation of the WS-Management protocol for remote management of Windows systems. It's the transport layer for PowerShell Remoting and Windows Admin Center. Configure with: `winrm quickconfig` (basic setup), Group Policy for enterprise deployment. Listeners are created for HTTP and HTTPS. `TrustedHosts` list controls which computers can connect without domain authentication.

Q10 What are Group Policy Objects (GPOs)?

Answer : Group Policy Objects are collections of settings that control the working environment of user accounts and computer accounts in an Active Directory domain. GPOs are linked to AD containers (sites, domains, OUs) and applied in order: Local > Site > Domain > OU (LSDOU). Processing order matters — later GPOs override earlier ones. Use `gpresult /r` to view applied policies, `gpupdate /force` to refresh immediately.

Q11 What is Group Policy processing order (LSDOU)?

Answer : GPOs are applied in this order: Local (stored on each computer), Site (linked to AD site), Domain (linked to AD domain), Organizational Unit (linked to OUs, from parent to child). Later policies override earlier ones for conflicting settings. Exceptions: No Override (Enforced) prevents child OUs from overriding, Block Inheritance prevents parent GPOs from applying. Computer policy applies at startup; user policy applies at logon.

Q12 What is the difference between Computer Configuration and User Configuration in GPO?

Answer : Computer Configuration: settings applied to computers regardless of who logs in — processed at startup, refreshed every 90 minutes. Affects OS settings, software installation for all users, and startup scripts. User Configuration: settings applied to users regardless of which computer they use — processed at logon, refreshed every 90 minutes. Affects desktop environment, software installation per-user, logon scripts, and folder redirection.

Q13 What is SYSVOL and what is it used for?

Answer : SYSVOL is a shared folder on each domain controller storing public files (Group Policy Templates, logon scripts) that must be replicated to all DCs. Located at `%SystemRoot%\SYSVOL\sysvol`. Replicated via DFS-R (modern) or FRS (legacy). SYSVOL contains: PolicyDefinitions (ADMX templates), {GUID} folders for each GPO (GPT — Group Policy Template), scripts folders, and domain folder. Critical for GPO application across the domain.

Q14 What is Desired State Configuration (DSC)?

Answer : PowerShell Desired State Configuration (DSC) is a management platform for declaring and enforcing server configurations. A DSC configuration script defines the desired state (e.g., service running, file exists, registry key value). The LCM (Local Configuration Manager) on each node applies and maintains the configuration. DSC modes: Push (directly apply), Pull (nodes fetch config from pull server). Integrates with Azure Automation State Configuration for cloud management.

Q15 What is the Local Configuration Manager (LCM)?

Answer : The LCM is the DSC engine running on every Windows node that processes DSC configurations. It reads the MOF (Managed Object Format) file and applies the desired state using DSC resources. Key LCM settings: RefreshMode (Push or Pull), ConfigurationMode (ApplyOnly, ApplyAndMonitor, ApplyAndAutoCorrect), RefreshFrequencyMins (how often to check for new config), RebootNodeIfNeeded. Configure with Set-DscLocalConfigurationManager.

Q16 What is Windows Server Update Services (WSUS)?

Answer : WSUS is a Windows Server role that enables IT administrators to manage and distribute Microsoft updates within their organization. WSUS downloads updates from Microsoft Update and distributes them to client computers. Features: approval workflow (approve/decline updates), computer groups for targeted deployment, synchronization scheduling, reporting on update compliance, and downstream WSUS servers for branch offices. Clients point to WSUS via Group Policy (WSUS server URL).

Q17 What are Windows Server licensing models?

Answer : Windows Server uses core-based licensing. Minimum: 16 cores per server (even if fewer physical cores). Standard edition: licenses 2 VMs per 16-core license (buy additional licenses for more VMs). Datacenter edition: unlimited VMs per licensed physical server. Client Access Licenses (CALs) required for each user or device accessing server. CAL types: Device CAL (one device, multiple users), User CAL (one user, multiple devices). CALs not needed for internet-facing services.

Q18 What is the Windows Server Backup feature?

Answer : Windows Server Backup (WSB) is a built-in backup solution for Windows Server. Features: bare-metal recovery, system state backup, volume backups, scheduled backups, VSS (Volume Shadow Copy Service) integration for application-consistent backups. Backs up to: local disk, network share, or removable media. For enterprise needs, use Microsoft Azure Backup, DPM (Data Protection Manager), or third-party solutions.

Q19 What is Volume Shadow Copy Service (VSS)?

Answer : VSS is a Windows technology that creates consistent point-in-time snapshots (shadow copies) of files and volumes, even when files are in use. VSS coordinates between requestors (backup software), writers (applications like SQL Server, Exchange), and providers (storage). This allows application-consistent backups without taking applications offline. Enable shadow copies on volumes via the Computer Management console or vssadmin command.

Q20 What is the Windows Event Log and key event channels?

Answer : Windows Event Log records system, security, and application events. Key channels: System (OS events – service failures, driver issues), Application (app events), Security (audit events – logon/logoff, resource access, policy changes), Setup (installation events). Modern apps use Applications and Services Logs. Event IDs 4624 (logon), 4625 (failed logon), 4740 (account lockout), 4776 (credential validation) are critical for security monitoring. Use Event Viewer or Get-WinEvent.

Q21 What is Windows Firewall with Advanced Security?

Answer : Windows Defender Firewall with Advanced Security is the host-based stateful firewall in Windows Server. Features: inbound and outbound rules, connection security rules (IPsec), profiles (Domain, Private, Public – applied based on network location), monitoring, and import/export of policies. Managed via the GUI (wf.msc), PowerShell (NetFirewallRule cmdlets), or Group Policy. Defense in depth layer between network firewall and server applications.

Q22 What is BitLocker and BitLocker To Go?

Answer : BitLocker is Windows full-volume encryption using AES-128 or AES-256. It protects data if a drive is physically removed. Requires TPM 2.0 (preferred) or startup key/PIN. Managed via Control Panel, PowerShell (Manage-BDE, BitLocker cmdlets), or Group Policy. BitLocker To Go extends BitLocker to removable drives (USB, external HDDs). BitLocker Network Unlock: automatically unlocks servers on trusted corporate network during startup. Manage recovery keys in Active Directory.

Q23 What is Windows Defender Antivirus in Windows Server?

Answer : Windows Defender Antivirus provides built-in malware protection in Windows Server 2016+. Features: real-time protection, cloud-delivered protection, automatic sample submission, tamper protection, and attack surface reduction rules. In Server Core installations, Defender is enabled by default. Manage via PowerShell (Get-MpComputerStatus, Update-MpSignature), Group Policy, or Microsoft Endpoint Configuration Manager. Consider Microsoft Defender for Endpoint for advanced EDR capabilities.

Q24 What are Windows Server Container technologies?

Answer : Windows Server supports two container isolation modes: Process isolation (Windows Containers) – shares host OS kernel, most efficient, fastest. Hyper-V isolation – each container runs in a lightweight VM, complete kernel isolation, more secure for multi-tenant scenarios. Use Docker Desktop or containerd as container runtime. Windows containers support Windows Server Core and Nano Server base images. Orchestrate with Kubernetes (AKS on Azure or kubeadm on-premises).

Q25 What is the Microsoft Deployment Toolkit (MDT)?

Answer : MDT is a free Microsoft tool for automating Windows OS and application deployment. It uses deployment shares, task sequences, and driver/application packages. MDT integrates with Windows Deployment Services (WDS) for network boot deployments (PXE boot). Task sequences define the deployment steps. MDT supports: Lite Touch Installation (LTI — minimal user interaction), Zero Touch Installation (ZTI — fully automated with System Center). Creates WIM images for deployment.

Q26 What is Windows Deployment Services (WDS)?

Answer : WDS is a Windows Server role for network-based OS installations using PXE (Pre-Boot Execution Environment). Clients boot from the network, connect to WDS server via TFTP/BINL, and receive OS images. WDS requires DHCP and DNS. Image types: Boot images (WinPE — used to start the deployment process), Install images (actual OS images deployed to computers). WDS integrates with MDT for task sequence-based deployments and automated installations.

Q27 What is System Center Configuration Manager (SCCM/MECM)?

Answer : Microsoft Endpoint Configuration Manager (MECM, formerly SCCM) is an enterprise tool for managing large numbers of Windows computers. Features: OS deployment (zero-touch), application deployment, software updates (integrates with WSUS), hardware/software inventory, compliance settings (desired configuration management), remote control, and reporting. Requires SQL Server. MECM co-management with Microsoft Intune bridges on-premises and cloud management.

Q28 What is Remote Desktop Services (RDS)?

Answer : Remote Desktop Services is a Windows Server role enabling users to access Windows desktops and applications remotely. Components: RD Session Host (RDSH — hosts sessions), RD Connection Broker (load balancing), RD Gateway (secure external access via HTTPS), RD Web Access (web portal), RD Licensing (manages RDS CALs), RD Virtualization Host (VDI). RDS can provide traditional session-based desktops or VDI (Virtual Desktop Infrastructure) with personal or pooled VMs.

Q29 What is the Windows Time Service (W32TM)?

Answer : The Windows Time Service maintains time synchronization across Windows computers using NTP (Network Time Protocol). In Active Directory, the PDC emulator (FSMO role) is the authoritative time source for the domain. Other DCs sync from PDC emulator; clients sync from their authenticating DC. Configure external NTP: `w32tm /config /manualpeerlist:time.windows.com /syncfromflags:manual /update`. Check sync status: `w32tm /query /status`. Accurate time is critical for Kerberos authentication (5-minute tolerance).

Q30 What is the Windows Registry?

Answer : The Windows Registry is a hierarchical database storing configuration settings for the OS, hardware, software, and users. Key hives: HKLM\SYSTEM (hardware config), HKLM\SOFTWARE (installed software settings), HKCU (current user settings), HKU (all user profiles), HKLM\SAM (local accounts). Edit via regedit (GUI), reg command (CLI), or PowerShell Get-ItemProperty/Set-ItemProperty. Back up before editing. Group Policy is the preferred way to manage registry settings at scale.

Q31 What is Windows Management Instrumentation (WMI)?

Answer : WMI is the Windows infrastructure for management data and operations. It provides a standardized interface for querying and configuring system components. WMI queries use WQL (WMI Query Language — similar to SQL): Get-WmiObject -Class Win32_OperatingSystem. PowerShell's Get-CimInstance (preferred modern approach) uses the same data. WMI is used by monitoring tools, SMS/SCCM, and management scripts. WMI repository stored at %SystemRoot%\System32\wbem.

Section 2: Active Directory Domain Services

Questions 31 – 60

Q31 What is Active Directory Domain Services (AD DS)?

Answer : AD DS is the core identity and authentication service in Windows Server networks. It stores information about network objects (users, computers, groups, printers) in a hierarchical, distributed database called the directory. AD DS provides: centralized authentication (Kerberos, NTLM), authorization (Group Policy, ACLs), single sign-on, policy management, and resource organization. The database is stored in NTDS.DIT on domain controllers.

Q32 What is a Domain Controller (DC)?

Answer : A Domain Controller is a Windows Server running the AD DS role that stores a copy of the AD database and handles authentication and authorization for the domain. Best practice: minimum 2 DCs per domain for redundancy. DCs replicate directory changes to each other. The first DC in a domain holds all 5 FSMO roles. Global Catalog servers (GC) store a partial read-only copy of all objects in the forest – required for universal group membership resolution and cross-domain queries.

Q33 What are FSMO roles?

Answer : Flexible Single Master Operation (FSMO) roles are domain controller roles that handle specific AD operations requiring a single authoritative source. Forest-wide (one per forest): Schema Master (manages schema changes), Domain Naming Master (adds/removes domains from forest). Domain-wide (one per domain): PDC Emulator (time sync, password changes, account lockouts, legacy compatibility), RID Master (issues Relative ID pools to DCs), Infrastructure Master (maintains cross-domain object references). Transfer with ntdsutil or AD Users and Computers.

Q34 What is the PDC Emulator FSMO role?

Answer : The PDC (Primary Domain Controller) Emulator is the most critical FSMO role, handling: authoritative time source for the domain (all computers sync time from it), immediate replication of password changes (client retries authentication against PDC emulator on failure), account lockout processing (receives lockout notifications from all DCs), Group Policy updates (edits go to PDC emulator by default), and legacy compatibility with pre-Windows 2000 clients. Seize this role first if DC fails.

Q35 What is the Global Catalog (GC)?

Answer : The Global Catalog is a domain controller that stores a full copy of all objects in its own domain and a partial (read-only) copy of all objects in all other domains in the forest. GC is required for: user principal name (UPN) logon resolution, universal group membership enumeration during logon (especially in multi-domain forests), Exchange Server (used for address book lookups), and cross-domain searches. The first DC in the forest is automatically a GC. Port 3268 (LDAP GC), 3269 (LDAP GC SSL).

Q36 What is an Active Directory Forest and Domain?

Answer : Forest: the top-level container of AD — a collection of one or more domains sharing a common schema, configuration partition, and Global Catalog. All domains in a forest automatically trust each other (transitive trusts). Domain: a partition of the forest with its own security boundary, OU structure, and domain-wide policies. The first domain is the Forest Root Domain. Child domains share the same namespace (child.parent.com). An AD forest can have multiple domain trees with different namespaces.

Q37 What is an Organizational Unit (OU)?

Answer : An Organizational Unit is a container within an AD domain for organizing users, computers, groups, and other OUs. OUs serve two primary purposes: delegation of administrative control (grant specific permissions within an OU without giving domain-wide admin rights) and Group Policy application (link GPOs to OUs for targeted policy application). Unlike domains, OUs are not security boundaries. Design OUs based on administrative requirements, not necessarily organizational chart.

Q38 What is an Active Directory Trust?

Answer : A trust is a relationship between domains or forests that allows authentication of users from one domain/forest in another. Trust types: Two-way (both domains trust each other), One-way (domain A trusts domain B — A's users can access B's resources). Trust transitivity: Transitive (trust extends through entire trust chain — all forest domains trust each other). Non-transitive (trust limited to two domains). External trusts are non-transitive; forest trusts are transitive within the trusted forest.

Q39 What are Active Directory Sites?

Answer : AD Sites represent physical network locations (e.g., branch offices) connected by reliable, high-bandwidth links. Sites contain subnets and are used to: control AD replication (replicate within site immediately; between sites on a schedule via site links), optimize authentication (clients use DCs in their own site first — site coverage), and control DFS namespace referrals. Site links define replication schedules, costs (lower = preferred), and replication intervals. Configure with AD Sites and Services MMC.

Q40 What is AD replication and what is the KCC?

Answer : AD replication synchronizes directory changes across all domain controllers. The Knowledge Consistency Checker (KCC) is an automated process that creates and maintains the replication topology (connection objects). Within a site: changes replicate within 15 seconds (urgent changes like password changes replicate immediately). Between sites: replication occurs on a schedule defined by site link objects (default 180 minutes). Replication uses Remote Procedure Call (RPC) for intra-site and either RPC or SMTP for inter-site.

Q41 What is the AD Database (NTDS.DIT)?

Answer : NTDS.DIT (NT Directory Services Directory Information Tree) is the Active Directory database file stored at %SystemRoot%\NTDS\NTDS.DIT on every domain controller. It uses the Extensible Storage Engine (ESE). The database contains all AD objects and their attributes. Maintenance operations: defragment with ntdsutil (offline defragmentation to reclaim space), check integrity with ntdsutil file integrity. The associated transaction log files (EDB*.log) enable crash recovery. Never copy NTDS.DIT while DC is running.

Q42 What is AD Recycle Bin?

Answer : AD Recycle Bin allows administrators to recover accidentally deleted Active Directory objects without restoring from backup. Deleted objects are preserved in a 'deleted' state with all attributes for a configurable period (Deleted Object Lifetime, default 180 days). Enable with: Enable-ADOptionalFeature -Identity 'Recycle Bin Feature'. Restore with: Restore-ADObject or AD Administrative Center. Requires Forest Functional Level 2008 R2 or higher. Once enabled, cannot be disabled.

Q43 What are Active Directory Functional Levels?

Answer : Functional levels define the minimum Windows Server version for all DCs in a domain/forest and unlock features. Domain Functional Level: all DCs must be at or above this version. Forest Functional Level: all domains must meet this level. Raising levels: one-way operation (cannot lower). Key features unlocked: 2008 R2 DFL = AD Recycle Bin, 2012 R2 = Protected Users security group, fine-grained password policy (PSOs) available from 2008 DFL. Use Get-ADDomain to check current level.

Q44 What is Fine-Grained Password Policy (FGPP)?

Answer : Fine-Grained Password Policies allow different password policies for different sets of users within the same domain, overcoming the limitation of one domain-wide password policy. Implemented via Password Settings Objects (PSOs) applied to users or global groups. PSO attributes match the Default Domain Policy password settings plus a precedence value (lower = higher priority). Requires 2008 DFL minimum. Manage with ADAC (Active Directory Administrative Center) or PowerShell `New-ADFineGrainedPasswordPolicy`.

Q45 What is Protected Users security group?

Answer : Protected Users is a security group in Windows Server 2012 R2+ that provides additional protection against credential theft. Members: cannot use NTLM, DES, or RC4 encryption (Kerberos AES only), cannot use unconstrained delegation, Kerberos TGT lifetime limited to 4 hours (not renewable), no credential caching. Use for: privileged accounts (domain admins, service accounts with high privilege). Warning: ensures Kerberos-only authentication — may break services requiring NTLM or long-lived tickets.

Q46 What are Active Directory Replication Conflicts?

Answer : Replication conflicts occur when the same object or attribute is modified on different DCs simultaneously. Resolution uses USN (Update Sequence Number) and Originating Write timestamps. Last-writer-wins for attribute-level conflicts: the change with the highest version number wins; if equal, later timestamp wins. Object-level conflicts: two objects created with the same relative distinguished name — one gets renamed with a CNF (conflict) prefix. Use `repadmin /showrepl` to check replication status and errors.

Q47 What is Kerberos authentication in Active Directory?

Answer : Kerberos is the default authentication protocol in AD environments. Process: 1) Client requests TGT from KDC (Key Distribution Center — DC) using AS-REQ. 2) KDC issues TGT encrypted with `krbtgt` account hash. 3) Client requests service ticket (ST) using TGT. 4) KDC issues ST encrypted with service account hash. 5) Client presents ST to service. Key components: `krbtgt` account (key for all tickets), TGT (Ticket Granting Ticket), ST (Service Ticket), SPN (Service Principal Name). Requires time sync within 5 minutes.

Q48 What is NTLM authentication and when is it still used?

Answer : NTLM (NT LAN Manager) is the legacy Windows challenge-response authentication protocol. Still used when: Kerberos not available (workgroup computers, IP address authentication, non-domain resources), or by legacy applications. NTLM authentication involves: client sends username, server sends challenge, client hashes challenge with password hash and sends response. NTLMv2 is the current version. Security risks: pass-the-hash attacks, no mutual authentication (server doesn't prove identity to client). Block NTLM via Group Policy where possible.

Q49 What is LDAP and how is it used with AD?

Answer : LDAP (Lightweight Directory Access Protocol) is the protocol used to query and modify Active Directory. AD listens on: port 389 (LDAP), 636 (LDAPS – LDAP over SSL), 3268 (GC), 3269 (GC SSL). LDAP queries use a distinguished name (DN) format: CN=JohnDoe,OU=Users,DC=contoso,DC=com. PowerShell uses LDAP filters: `Get-ADUser -LDAPFilter '(objectCategory=person)'`. LDAP is used by non-Windows applications (Java apps, Linux systems) to query AD. Enable LDAPS by installing a certificate on DCs.

Q50 What is AD LDS (Lightweight Directory Services)?

Answer : AD LDS is a standalone LDAP directory service that doesn't require a full AD domain. Uses: storing application-specific directory data, providing LDAP authentication for internet-facing applications without exposing AD, supporting multiple independent directory instances on one server. AD LDS data stores are separate from AD DS. Applications can use AD LDS without AD domain membership. Managed with ADSI Edit, LDP, or PowerShell. Commonly used with ADFS for claims-based identity.

Q51 What is Active Directory Certificate Services (AD CS)?

Answer : AD CS is a Windows Server role for creating and managing a Public Key Infrastructure (PKI). It issues digital certificates for: user and computer authentication (smart card logon), encrypting file system (EFS), SSL/TLS certificates for web servers, email signing and encryption, and code signing. Components: Certification Authority (CA), Online Responder (OCSP), Certificate Enrollment Web Services, NDES (Simple Certificate Enrollment Protocol for network devices). Two-tier hierarchy: Offline Root CA + Online Issuing CA.

Q52 What is AD FS (Active Directory Federation Services)?

Answer : AD FS is a Windows Server role providing claims-based authentication for federated identity scenarios — single sign-on across organizational boundaries. It implements SAML 2.0, WS-Federation, and OAuth 2.0/OpenID Connect standards. Common uses: SSO with Microsoft 365/Azure AD (federated identity), partner extranet access (B2B), and web application access without VPN. AD FS uses relying party trusts (applications trusting AD FS) and claims provider trusts (identity sources). Being replaced by Azure AD for most scenarios.

Q53 What is AD RMS (Rights Management Services)?

Answer : AD RMS provides information protection by enforcing policies on documents and emails. Organizations define usage rights (read, print, forward, copy) on Office documents and emails. Users can open RMS-protected content only if they have the required rights. Content remains protected even if shared externally. Requires: AD RMS server role, AD RMS client on user devices, SQL Server. Being replaced by Azure Information Protection (AIP) / Microsoft Purview for cloud-based protection.

Q54 What is the difference between security groups and distribution groups?

Answer : Security Groups: used for assigning permissions to resources (NTFS permissions, share permissions, GPO filtering) AND as email distribution lists. Types: Domain Local, Global, Universal. Distribution Groups: used only for email distribution (Exchange/Microsoft 365) — cannot be used for resource permissions. Always use security groups for resource access control. Universal security groups store membership in Global Catalog — avoid large memberships that change frequently.

Q55 What are the Active Directory group scopes?

Answer : Domain Local: members from any domain in forest + other forests; can assign permissions only within same domain. Best for resource permissions. Global: members from same domain only; can be used in any domain in forest. Best for organizing users by department/function. Universal: members from any domain in forest; stored in Global Catalog; used in any domain. Best for cross-domain groups but avoid frequent membership changes (triggers GC replication). Rule of thumb: AGDLP (Accounts □ Global □ Domain Local □ Permissions).

Q56 What is the AGDLP strategy?

Answer : AGDLP is the Microsoft recommended strategy for managing access permissions in AD: Accounts (user accounts) added to Global groups (organized by function/department), Global groups added to Domain Local groups (organized by resource/permissions), Domain Local groups assigned Permissions on resources. Benefits: easier administration (change access by modifying global group membership), flexibility (domain local groups can span resources, global groups can nest in multiple DLs), and scalability. Extend to AGUDLP for universal groups in multi-domain forests.

Q57 What are Service Principal Names (SPNs)?

Answer : An SPN uniquely identifies a service instance for Kerberos authentication. Format: serviceclass/host:port/servicename (e.g., HTTP/webserver.contoso.com). When a client authenticates to a service using Kerberos, it requests a ticket for the service's SPN. SPNs must be unique within the forest. Register/check with: setspn -L servername (list), setspn -A HTTP/server account (add). Duplicate or missing SPNs cause Kerberos authentication failures. AD service accounts need SPNs registered.

Q58 What is Managed Service Accounts (MSA) and gMSA?

Answer : Managed Service Accounts (MSA) and Group Managed Service Accounts (gMSA) are domain accounts for services that automatically manage their own passwords. MSA: for services on a single server, passwords auto-rotated every 30 days, no password management needed. gMSA: extended MSA for services running on multiple servers (web farms, NLB clusters), password managed by Key Distribution Service (KDS). Create gMSA: New-ADServiceAccount. Install: Install-ADServiceAccount. Use instead of regular service accounts with static passwords.

Q59 What is AD backup and recovery?

Answer : AD DS backup uses Windows Server Backup to perform System State backup (includes NTDS.DIT, SYSVOL, registry, boot files). For authoritative restore (restore deleted objects): boot into DSRM (Directory Services Restore Mode), restore System State, mark objects as authoritative with ntdsutil 'authoritative restore'. Non-authoritative restore: normal restore — AD replication will update the DC with current data from other DCs. Snapshot-based backup (VMs) requires AD-aware backup solutions to avoid USN rollback.

Q60 What is Azure AD Connect and hybrid identity?

Answer : Azure AD Connect synchronizes on-premises AD identities to Azure Active Directory (Entra ID) for hybrid identity scenarios. Sync methods: Password Hash Synchronization (PHS — hash of hash synced to Azure AD), Pass-through Authentication (PTA — validates against on-prem AD in real time), Federation with AD FS (delegates to on-prem AD FS). Enables: same credentials for on-prem and cloud, single sign-on to Microsoft 365 and Azure, conditional access based on on-prem attributes. Install on a member server, not a DC.

Q61 What is Read-Only Domain Controller (RODC)?

Answer : An RODC holds a read-only copy of the AD database — no changes can be written directly to it. Changes sync from writable DCs only. Use cases: branch offices with poor physical security (if RODC is stolen, no writable copy of AD is exposed), untrusted network zones. Features: filtered attribute set (sensitive attributes not replicated to RODC), credential caching (only allowed accounts' credentials cached), administrator role separation (local admin without domain admin rights). Password Replication Policy controls which accounts' passwords are cached.

Section 3: DNS & DHCP

Questions 61 – 90

Q61 What is DNS Server role in Windows Server?

Answer : The DNS (Domain Name System) Server role in Windows Server resolves hostnames to IP addresses and vice versa. Critical for AD DS – domain controllers register SRV records in DNS so clients can find DCs. Windows DNS Server supports: Primary zones (authoritative, read-write), Secondary zones (read-only replica from primary), Stub zones (stores NS records only – used for zone delegation), and Active Directory Integrated zones (stored in AD, replicated via AD replication). DNS uses UDP port 53 (queries) and TCP port 53 (zone transfers, large responses).

Q62 What are Active Directory Integrated DNS zones?

Answer : AD Integrated zones store DNS zone data in the AD database instead of a flat file. Benefits: secure dynamic updates (only domain members can register records), multi-master replication (all DCs with DNS can update the zone – no primary/secondary distinction), replication uses AD replication topology (more efficient), automatic replication when new DCs are added. Zone data is stored in AD Application Partitions: DomainDNSZones (replicated to DCs in same domain) or ForestDNSZones (replicated to all DCs in forest).

Q63 What are DNS record types?

Answer : Common DNS record types: A (maps hostname to IPv4 address), AAAA (maps hostname to IPv6 address), CNAME (alias for another hostname), MX (mail exchange server for a domain), PTR (reverse lookup – IP to hostname), NS (name server for a zone), SOA (Start of Authority – zone settings including serial number, refresh, retry intervals), SRV (service location – used by AD, Kerberos, SIP), TXT (text records – used for SPF, DKIM, domain verification). AD requires: _msdcs SRV records, _kerberos, _ldap.

Q64 What is DNS dynamic update?

Answer : DNS dynamic update allows DNS clients and DHCP servers to automatically register and update DNS records without manual intervention. Types: Non-secure (any client can register – not recommended), Secure (only authenticated domain members can register – requires AD integrated zones). Configure in DNS zone properties. DHCP servers can register records on behalf of legacy clients (Secure Updates on Behalf of DHCP clients). Dynamic update uses the DDNS protocol (DNS UPDATE messages). Records have scavenging timestamps to detect stale records.

Q65 What is DNS scavenging?

Answer : DNS scavenging automatically removes stale DNS resource records that are no longer needed. Records have a timestamp when dynamically registered. Scavenging process: No-refresh interval (record cannot be refreshed — prevents timestamp updates too frequently, default 7 days), Refresh interval (record can be refreshed — if not refreshed by end of this period, eligible for scavenging, default 7 days). Enable scavenging on the DNS server and on each zone. Run immediately: `dnscmd /startscavenging`. Stale records cause authentication and connectivity issues.

Q66 What is DNS forwarding and conditional forwarding?

Answer : DNS forwarding: when a DNS server can't resolve a query, it forwards to another DNS server (forwarder) instead of recursing to root hints. Configure: DNS Manager > server properties > Forwarders. Conditional forwarding: forward queries for specific domain names to specific DNS servers. Example: forward all *.contoso.com queries to contoso.com DNS servers. Used in: internal DNS forwarding to ISP, split DNS scenarios, AD forest integration. Forwarder IP configured per domain name. Stored as stub zones or forwarder objects.

Q67 What is DNS split-brain (split-horizon)?

Answer : Split-brain DNS provides different DNS responses for the same hostname based on the source of the query — internal users get private IP, external users get public IP. Implementation: maintain two DNS zones with the same name — one for internal (private IPs), one for external (public IPs). Azure Private DNS zones + Azure DNS public zones implement this for hybrid scenarios. Also implemented with Windows DNS views or separate DNS servers for internal vs DMZ networks.

Q68 What is DNS Security Extensions (DNSSEC)?

Answer : DNSSEC adds cryptographic signatures to DNS records to ensure authenticity and integrity. Protects against DNS cache poisoning and man-in-the-middle attacks. Windows Server DNS supports DNSSEC: sign zone with KSK (Key Signing Key) and ZSK (Zone Signing Key), distribute trust anchors, and configure NRPT (Name Resolution Policy Table) to require DNSSEC for specific namespaces. DNS responses include RRSIG (resource record signature) and DNSKEY records. Complex to implement; consider Azure DNS with DNSSEC where available.

Q69 What is the DNS Name Resolution Policy Table (NRPT)?

Answer : NRPT is a table of rules that determines how DNS queries are handled based on the domain namespace. It can: enforce DNSSEC validation for specific namespaces, route queries for specific domains to specific DNS servers, and configure DirectAccess/VPN-split DNS behavior. Configured via Group Policy (Computer Configuration > Windows Settings > Name Resolution Policy). Example: all queries for contoso.com use internal DNS server 10.0.0.1 while other queries use default DNS. Takes precedence over standard DNS client settings.

Q70 What is the DHCP Server role?

Answer : DHCP (Dynamic Host Configuration Protocol) automatically assigns IP addresses and network configuration (subnet mask, default gateway, DNS server, WINS server) to network clients. Windows Server DHCP features: scopes (ranges of IP addresses), exclusions (reserved addresses within scope), reservations (static assignment by MAC address), superscopes (multiple scopes on one subnet), multicast scopes, failover (active-active or active-passive with another DHCP server), and DHCP policies for conditional option assignment.

Q71 What is a DHCP scope and its components?

Answer : A DHCP scope defines a range of IP addresses available for dynamic assignment on a subnet. Components: IP address range (start and end IP), Subnet mask, Exclusion range (addresses within scope not to assign – for static IPs), Reservations (specific IP always given to specific MAC address), Lease duration (how long client keeps the IP – default 8 days), Scope options (gateway, DNS server, DNS domain, etc.), Activation status (must be activated to assign addresses).

Q72 What is DHCP failover?

Answer : DHCP failover allows two DHCP servers to share responsibility for a scope, providing redundancy. Two modes: Hot Standby (one active, one standby – standby takes over if active fails; active serves 100% of scope, typically 5% reserved for standby), Load Sharing (both servers active simultaneously, each serving a percentage of clients – default 50/50). Failover partners communicate on port 647. Configure in DHCP Manager. Requires Windows Server 2012 or later on both servers.

Q73 What is DHCP split scope (pre-2012 failover)?

Answer : DHCP split scope is the legacy failover method (pre-Windows Server 2012) where the same scope IP range is configured on two DHCP servers, each serving a portion: typically 80/20 split – primary serves 80% of addresses, secondary serves 20%. The scope on each server has exclusions preventing assignment of the other server's addresses. Both servers are active simultaneously. Limitation: no state sharing between servers – clients may get different IPs from each server.

Q74 What are DHCP options?

Answer : DHCP options provide network configuration to clients along with an IP address. Common options: Option 003 (Router/Default Gateway), Option 006 (DNS Servers), Option 015 (DNS Domain Name), Option 044 (WINS/NBNS Servers), Option 046 (NetBIOS Node Type), Option 066 (TFTP Server – for PXE boot), Option 067 (Bootfile Name – for PXE). Options can be set at server level (all scopes), scope level (specific subnet), policy level (conditional), or reservation level (specific client).

Q75 What is DHCP authorization?

Answer : DHCP authorization prevents unauthorized (rogue) DHCP servers from operating in an AD domain environment. Windows Server DHCP servers must be authorized in AD by a domain/enterprise administrator. Non-authorized DHCP servers shut down automatically if they detect an authorized DHCP server. Configuration: DHCP Manager > right-click server > Authorize. Uses AD to store list of authorized DHCP server IP addresses. Only applies to Windows DHCP servers in AD domains – Linux or unauthorized Windows servers are not controlled.

Q76 What is DNS and DHCP integration?

Answer : DHCP integrates with DNS for automatic DNS registration. When DHCP assigns an IP, it can register the A record (client's hostname to IP) and PTR record (IP to hostname) in DNS on the client's behalf. Configure in DHCP scope properties > DNS tab. For Windows clients: client registers own A record, DHCP registers PTR. For legacy (non-Windows) clients: DHCP registers both records. Enable Secure Dynamic Updates for AD-integrated zones – DHCP server registers records as the client's name in DNS. Prevents orphaned DNS records from DHCP reassignments.

Q77 What is the ipconfig command for DHCP?

Answer : ipconfig is the primary command-line tool for IP configuration in Windows. Key DHCP-related commands: ipconfig /all (show full IP configuration including DHCP server, lease obtained/expires), ipconfig /release (release DHCP lease – IP returned to pool), ipconfig /renew (request new IP from DHCP), ipconfig /flushdns (clear DNS cache), ipconfig /registerdns (re-register DNS records with DNS server), ipconfig /displaydns (show DNS cache contents). Run with admin rights for release/renew.

Q78 What is the difference between primary and secondary DNS zones?

Answer : Primary DNS zone: the authoritative, read-write copy of a zone. Zone transfers copy data FROM the primary to secondary zones. Standard primary: stored as a text file on the DNS server. AD-integrated primary: stored in AD database, replicated via AD. Secondary DNS zone: a read-only copy of the zone received from the primary (or another secondary) via zone transfer. Cannot be modified directly. Provides redundancy and load distribution for DNS resolution. Zone transfer uses TCP port 53 (AXFR = full transfer, IXFR = incremental transfer).

Q79 What is DNS root hints vs DNS forwarding?

Answer : Root Hints: list of authoritative DNS servers for the root zone (.) — used when a DNS server needs to resolve a name it's not authoritative for, starting from the top of the DNS hierarchy (iterative queries). Default root hints point to internet root servers. Forwarding: instead of using root hints and iterative queries, forward unknown queries to a specified DNS server (resolver). Internal DNS servers typically forward to ISP or public DNS (8.8.8.8). If forwarding fails and 'Use root hints if no forwarders available' is enabled, root hints are used as fallback.

Q80 What is NSLOOKUP and how do you use it for DNS troubleshooting?

Answer : NSLOOKUP is a command-line tool for querying DNS servers. Common uses: nslookup hostname (resolves hostname using default DNS server), nslookup hostname dns-server (query specific DNS server), nslookup -type=MX domain (query for mail records), nslookup -type=SRV _kerberos._tcp.domain (query SRV records for Kerberos), set debug (verbose output). For interactive mode: type 'nslookup' then commands. Also use: Resolve-DnsName (PowerShell), dig (Windows via toolset).

Q81 What is DNS delegation?

Answer : DNS delegation is the process of assigning authority for a subdomain to a different DNS server/zone. The parent zone contains NS records pointing to the authoritative DNS servers for the child zone, and glue records (A records for the name servers). Example: contoso.com delegates east.contoso.com to 10.10.10.1 — queries for *.east.contoso.com are referred to that server. Used for: dividing large DNS namespaces, geographic distribution of DNS management, and Active Directory child domains.

Q82 What is Reverse DNS (PTR records)?

Answer : Reverse DNS resolves IP addresses to hostnames. PTR (pointer) records stored in reverse lookup zones use in-addr.arpa format (IPv4) or ip6.arpa (IPv6). Zone name is the reversed network portion: for 192.168.1.x, the reverse zone is 1.168.192.in-addr.arpa. PTR record: 10.1.168.192.in-addr.arpa □ server1.contoso.com. Used by: email servers (spam filtering), network monitoring, and logging (correlate IPs to names). Create reverse lookup zones in DNS Manager and configure DHCP to register PTR records.

Q83 What are DNS stub zones?

Answer : A stub zone contains only the NS records, SOA record, and A records (glue records) for the authoritative name servers of another zone — not the complete zone data. Stub zones are used to: maintain awareness of authoritative DNS servers for a zone without full secondary zone replication, automatically update NS records when they change in the primary zone, and improve performance by avoiding unnecessary DNS recursion. AD-integrated stub zones replicate via AD. Used in AD forest root configurations and trust scenarios.

Q84 What is the DHCP Database backup and restore?

Answer : The DHCP database is automatically backed up every 60 minutes to %SystemRoot%\System32\DHCP\Backup. Restore options: Restore-DhcpServer PowerShell cmdlet for restoring server configuration and scope data, or copy backup files and use reconciliation. To move DHCP server: Export-DhcpServer (export configuration to XML), Import-DhcpServer (import on new server). After restore, reconcile scopes (DHCP Manager > scope > Reconcile) to fix inconsistencies between the DHCP database and registry.

Q85 What is IPv6 in Windows Server DNS and DHCP?

Answer : Windows Server DNS fully supports AAAA records (IPv6 address records) alongside A records. Windows Server DHCP supports DHCPv6 for dynamically assigning IPv6 addresses — configure IPv6 scopes in DHCP Manager. IPv6 also uses SLAAC (Stateless Address Autoconfiguration) — devices auto-configure using router advertisements (no DHCP needed). AD DS operates over IPv6. By default, Windows prefers IPv6 over IPv4 when both are available. Disable IPv6 only if required for compatibility, not as general practice.

Q86 What is the DNS debug log?

Answer : DNS debug logging records detailed DNS query and response information for troubleshooting. Enable in DNS Manager > server properties > Debug Logging tab. Configurable: log queries, responses, specific transports (UDP/TCP), packet direction, specific IP address filters, log file path and size. Creates detailed log at %SystemRoot%\System32\DNS\DNS.log. Warning: verbose logging impacts DNS server performance — enable only for troubleshooting and disable when done. Prefer DNS event logging (Event Viewer) for production monitoring.

Q87 What is DNSLint?

Answer : DNSLint is a Microsoft diagnostic tool (dnslint.exe from Support Tools) for verifying DNS records related to Active Directory and diagnosing DNS replication problems. Common usage: dnslint /ad (verify AD DNS registration), dnslint /s dnsserver (test DNS server), dnslint /ql query-list-file (test DNS queries). While DNSLint is older, it's still used for AD DNS verification. Modern alternatives: dcdiag /test:dns, nslookup, Resolve-DnsName. Dcdiag /test:dns is the preferred method for AD DNS validation.

Q88 What is DHCP Class Options?

Answer : DHCP class options allow DHCP to deliver different configuration options based on client characteristics. User-defined classes: clients specify a class ID (Option 77 – User Class) and receive class-specific options. Vendor classes: based on DHCP Option 60 (vendor class identifier) – assign options to specific OS or device types (Windows PCs vs IP phones vs printers). Example: assign different DNS servers to VoIP phones vs workstations. Define classes in DHCP Manager > Define User Classes or Define Vendor Classes.

Q89 What is DHCP Policies (Windows Server 2012+)?

Answer : DHCP policies (introduced in Windows Server 2012) allow conditional assignment of IP addresses and options based on client characteristics: hardware type, manufacturer, MAC address, fully qualified domain name, client ID, relay agent information, and user class. Policies have priority (lower = higher priority) and can set: IP address range within scope, TTL, and any DHCP options. More powerful than class-based options – can combine multiple conditions with AND/OR logic.

Q90 What is DNS aging and scavenging versus manual record management?

Answer : DNS aging/scavenging automates removal of stale dynamically registered records. Manual record management: administrators create, modify, delete records manually – appropriate for static infrastructure (servers, printers). For dynamic environments (workstations, laptops), aging/scavenging is essential to prevent DNS pollution with stale records. Setting: enable aging on both the zone and server level, configure No-Refresh and Refresh intervals (default 7 days each). Statically created records are not scavenged unless timestamps are enabled manually.

Section 4: Hyper-V & Virtualization

Questions 91 – 120

Q91 What is Hyper-V and what are its editions?

Answer : Hyper-V is Microsoft's hardware virtualization platform built into Windows Server (role) and available as a free standalone hypervisor (Hyper-V Server). It's a Type 1 (bare-metal) hypervisor — runs directly on hardware, with the management OS running as a privileged VM (root partition). Hyper-V is available in: Windows Server Standard (2 VMs), Datacenter (unlimited VMs), Windows 10/11 Pro/Enterprise (Hyper-V for development), and free Hyper-V Server (headless, limited to Hyper-V management).

Q92 What are the Hyper-V virtual switch types?

Answer : Hyper-V has three virtual switch types: External switch — connects VMs to physical network through a physical NIC, VMs can communicate with external network and host. Internal switch — connects VMs to each other and to the Hyper-V host only; no connection to external physical network. Private switch — connects VMs to each other only; host cannot communicate with VMs on this switch. Create with Hyper-V Manager or New-VMSwitch PowerShell cmdlet.

Q93 What is VM Generation 1 vs Generation 2 in Hyper-V?

Answer : Generation 1 (legacy): emulates older hardware (IDE controller, BIOS, legacy network adapter), supports broader OS compatibility including older Windows and Linux, PXE boot via legacy adapter. Generation 2 (modern): UEFI firmware (no BIOS emulation), Secure Boot support, SCSI boot, synthetic-only devices (faster performance, no emulation overhead), smaller boot disk, supports Windows Server 2012+, and modern Linux distributions. Choose Gen 2 for supported OSes; use Gen 1 only for compatibility with older systems.

Q94 What is Dynamic Memory in Hyper-V?

Answer : Dynamic Memory allows Hyper-V to automatically adjust the amount of RAM assigned to a running VM based on workload demands. Key settings: Startup RAM (initial allocation when VM starts), Minimum RAM (floor — VM won't go below this), Maximum RAM (ceiling — VM won't exceed this), Memory Buffer % (percentage above current demand to keep available), Memory Weight (priority for memory allocation when host memory is constrained). The Hyper-V Dynamic Memory driver in the VM communicates memory pressure to the hypervisor. Requires Hyper-V Integration Services.

Q95 What are Hyper-V Integration Services?

Answer : Integration Services (IS) are a set of drivers and services that improve VM performance and provide enhanced functionality when installed in the guest OS. Services: Heartbeat (host detects if VM is alive), Time Synchronization (sync VM clock with host), Data Exchange (share key-value pairs between host and VM), Backup (VSS integration for application-consistent backup), Guest Shutdown (graceful shutdown from host), Guest Services (copy files into VM from host). Integration Services come with Windows Server. For Linux: linux-hyperv drivers in kernel.

Q96 What is VHD vs VHDX?

Answer : VHD (Virtual Hard Disk): older format, maximum size 2 TB, used by Hyper-V and Virtual PC. VHDX (Virtual Hard Disk Extended): modern format, maximum size 64 TB, larger block size (better performance for large VMs), built-in protection against data corruption during power failures (log file), supports 4 KB native sector disks, and trimming/unmap support for thin provisioning on shared storage. Always use VHDX for new VMs. Convert VHD to VHDX: Convert-VHD cmdlet or Hyper-V Manager Edit Disk wizard.

Q97 What is the difference between Fixed, Dynamic, and Differencing VHDs?

Answer : Fixed: allocated to full size at creation — best I/O performance, predictable storage consumption. Dynamic (thin): starts small and expands as data is written — space efficient but slower performance and fragmentation risk. Differencing: stores only changes relative to a parent disk — used for test environments and VDI (child VMs share parent image). For production VMs: use fixed disks for performance-critical workloads. Dynamic disks are acceptable for dev/test. Compact dynamic VHDs periodically to reclaim space.

Q98 What is Live Migration in Hyper-V?

Answer : Live Migration moves a running VM from one Hyper-V host to another with no downtime (brief network hiccup). Requirements: hosts in same cluster (for clustered storage — Cluster Shared Volumes) or Shared Nothing Live Migration (requires sufficient memory on destination, not shared storage), compatible processors (same manufacturer, enable processor compatibility mode for different generations), network configured for live migration. Process: VM state and memory are copied to destination while VM keeps running on source; when sync complete, VM state switches to destination, source VM shuts down.

Q99 What is Storage Migration in Hyper-V?

Answer : Storage Migration moves a VM's storage (VHDs/VHDXs) to a different location while the VM is running. Can move: VHD files, VM configuration files, or both. Can be to a different local disk, different CSV, or different SMB share. No cluster membership required (unlike Live Migration for cluster scenarios). Process: destination storage is created as a differencing disk, changes are synced, and switchover happens. Used for: storage rebalancing, migrating to faster storage, and consolidating storage.

Q10 0 What is Hyper-V Replica?

Answer : Hyper-V Replica provides asynchronous VM replication to another Hyper-V host or cluster for disaster recovery — without shared storage. Replication occurs over HTTP (port 80) or HTTPS (port 443). Replication frequency: 30 seconds, 5 minutes, or 15 minutes. Supports extended replication (primary □ replica □ extended replica for 3-site DR). Recovery points: up to 24 additional hourly recovery points. Failover types: Planned Failover (graceful, no data loss), Test Failover (non-disruptive — creates test VM), Unplanned Failover (disaster scenario — possible data loss of last replication interval).

Q10 1 What are Hyper-V Snapshots (Checkpoints)?

Answer : Checkpoints (formerly called Snapshots) capture the state of a VM at a point in time, allowing rollback. Types: Standard checkpoint — captures VM state (memory, CPU) and storage at a moment; quick operation but can cause storage inconsistencies for active databases. Production checkpoint — uses VSS (or Linux fsfreeze) for application-consistent checkpoints; doesn't capture running memory state; safer for production use. Apply checkpoint: VM rolls back to that state. Delete checkpoint: merges differencing disk into parent. Never use checkpoints in production for long periods.

Q10 2 What is Cluster Shared Volumes (CSV)?

Answer : CSV is a file system that allows multiple nodes in a Windows Server Failover Cluster to simultaneously read and write to the same NTFS volume. This enables Live Migration without storage migration — VMs on CSV can be live migrated between any cluster node. CSV volumes appear at C:\ClusterStorage\Volume1, 2, etc. CSV uses coordinator node concept (one node coordinates metadata operations). CSV cache improves read performance. Required for: Scale-Out File Server, VMs with Live Migration, SQL Server FCI with Hyper-V.

Q10 3 What is Shielded VMs?

Answer : Shielded VMs are Hyper-V VMs with strong security features protecting against unauthorized access — even from compromised Hyper-V hosts or fabric administrators. Components: vTPM (virtual TPM — enables BitLocker inside VM), HGS (Host Guardian Service — validates that hosts are compliant before running shielded VMs), KeyProtector (VHDX encryption keys protected by HGS keys), Shielding Data File (.pdk — contains owner certificate and answer files). Fabric admin cannot view, copy, or modify VM contents. Best for multi-tenant hosting and compliance scenarios.

Q10 4 What is the Host Guardian Service (HGS)?

Answer : HGS is a Windows Server role that provides attestation and key protection services for Shielded VMs. Two attestation modes: Admin-trusted (AD group membership determines trusted hosts), TPM-trusted (cryptographic measurement of host firmware, drivers, and boot configuration using TPM — more secure). HGS performs attestation when a Hyper-V host requests permission to run a shielded VM. If attestation passes, HGS releases the key protector allowing the VM to start. HGS should be deployed as a highly available cluster.

Q10 5 What is VMware to Hyper-V migration?

Answer : Moving VMs from VMware ESXi/vSphere to Hyper-V. Methods: Microsoft Virtual Machine Converter (MVMC — deprecated, superseded by other tools), manual conversion (export VMware VM, convert VMDK to VHDX using Convert-VHD or StarWind V2V, create new Hyper-V VM), and third-party tools (Zerto, Veeam, StarWind V2V Converter). Steps: quiesce VMware VM, export or convert disk, create Hyper-V VM with appropriate generation, attach converted VHDX, install Hyper-V Integration Services, remove VMware tools. Azure Migrate supports automated VMware to Azure (Hyper-V based) migration.

Q10 6 What is SR-IOV in Hyper-V networking?

Answer : Single Root I/O Virtualization (SR-IOV) allows a physical NIC to present multiple virtual functions (VFs) directly to VMs, bypassing the hypervisor virtual switch for near-native network performance. Benefits: lower latency, higher throughput, reduced CPU overhead for networking. Requirements: SR-IOV capable NIC, BIOS/UEFI support, Gen 2 VM. Configure: `Set-VMNetworkAdapter -VMName VM1 -IovEnabled $true`. Limitation: VMs with SR-IOV cannot be live migrated (VF is bound to physical hardware). Use for latency-sensitive workloads (HPC, storage replication).

Q10 7 What is NIC Teaming in Windows Server?

Answer : NIC Teaming (Load Balancing and Failover – LBFO) combines multiple physical NICs into a single logical interface for fault tolerance and/or bandwidth aggregation. Modes: Switch Independent (each NIC can connect to different switches), Switch Dependent (NICs connect to same switch – LACP/802.3ad). Load balancing: Hyper-V Port, Address Hash (distributes by MAC/IP/port hash), Dynamic. Teams appear as single network adapter. Configure via Server Manager > Local Server > NIC Teaming or New-NetLbfoTeam PowerShell. Note: Microsoft has deprecated LBFO in Server 2022, recommending Switch Embedded Teaming (SET) instead.

Q108 What is Switch Embedded Teaming (SET)?

Answer : SET is a NIC teaming solution integrated directly into the Hyper-V virtual switch, replacing the older LBFO NIC teaming for virtualized environments. SET supports RDMA (Remote Direct Memory Access) and SR-IOV, which LBFO does not support with Hyper-V. Create SET team: `New-VMSwitch -EnableEmbeddedTeaming $true -NetAdapterName 'NIC1','NIC2'`. SET provides: fault tolerance (automatic failover), bandwidth aggregation across multiple physical NICs. Required for SDN (Software Defined Networking) deployments. Maximum 8 NICs per SET team.

Q109 What is RDMA in Windows Server?

Answer : Remote Direct Memory Access (RDMA) allows network data to be transferred directly between application memory on different computers without involving the host CPU or OS – extremely low latency and high throughput. Used for: iWARP (TCP/IP based RDMA), RoCE (RDMA over Converged Ethernet), InfiniBand. SMB Direct uses RDMA for high-performance file sharing (Storage Spaces Direct, Scale-Out File Server). Configure with: `Get-NetAdapterRdma`, `Enable-NetAdapterRdma`. Requires RDMA-capable NICs (Mellanox, Intel, Cavium/Marvell).

Q110 What is Nested Virtualization?

Answer : Nested Virtualization allows running Hyper-V within a Hyper-V VM – a VM acting as a Hyper-V host running its own VMs. Use cases: testing Hyper-V features, lab environments, Azure Kubernetes Service on Azure VMs. Enable: `Set-VMProcessor -VMName VM1 -ExposeVirtualizationExtensions $true` (must be done while VM is off). The parent VM must run Windows Server 2016+ or Windows 10 Anniversary Update+. Performance is lower than native. Azure supports nested virtualization on specific VM SKUs (v3, Ev3, Dv3 and higher).

Q111 What is Discrete Device Assignment (DDA)?

Answer : DDA allows a VM to have direct, exclusive access to a physical hardware device (PCIe device — GPU, NVMe, network card) bypassing the hypervisor, providing near-native hardware performance. Use cases: GPU compute workloads (ML training), high-speed storage (NVMe direct to VM). The host loses access to the assigned device. Requirements: hardware IOMMU support, DDA-compatible device. Dismount device: Dismount-VMHostAssignableDevice, then Add-VMAssignableDevice to VM. Remove device from host and VM before adding to another VM.

Q11 2 What is the Hyper-V Manager vs System Center Virtual Machine Manager?

Answer : Hyper-V Manager: free built-in GUI for managing Hyper-V on individual hosts or small number of servers. Limited to Hyper-V management only. System Center Virtual Machine Manager (SCVMM/VMM): enterprise management for Hyper-V, VMware, and Xen virtualization platforms. Features: library management (VM templates, VHDs), fabric management (networks, storage, hosts), self-service portal, capacity planning, and automation via PowerShell and runbooks. Part of System Center suite. Windows Admin Center is a modern alternative for Hyper-V management without System Center cost.

Q11 3 What is Azure Stack HCI?

Answer : Azure Stack HCI is a hyperconverged infrastructure (HCI) platform for running virtualized workloads on-premises. It combines Hyper-V for compute, Storage Spaces Direct for software-defined storage, and Software-Defined Networking. It's delivered as a subscription service billed per core per month. Features: native Azure integration (Azure Arc), Azure Monitor, Azure Backup, Azure Site Recovery, and access to Azure Hybrid services. Requires validated hardware from Microsoft partners. Minimum 2 nodes, maximum 16 nodes per cluster. Ideal for replacing aging hardware while maintaining Azure-consistent management.

Q11 4 What is Storage QoS in Hyper-V?

Answer : Storage Quality of Service in Hyper-V allows administrators to define minimum and maximum IOPS (Input/Output Operations Per Second) for VM storage. This prevents 'noisy neighbor' scenarios where one VM consuming all storage bandwidth affects others. Configure per-VHD or per-VM. Policies created in SCVMM or Windows PowerShell: New-StorageQosPolicy. Minimum IOPS ensures performance floor; maximum IOPS enforces ceiling. Metrics visible in Performance Monitor and Hyper-V WMI. Requires Windows Server 2012 R2+.

Q11 5 What are VM checkpoints best practices?

Answer : Best practices for checkpoints: Use Production Checkpoints for production VMs (application-consistent via VSS). Never leave checkpoints in place long-term – differencing disks grow and impact performance. Do not use checkpoints for database VMs (SQL, Exchange) as substitutes for proper application backups – use application-aware backup. Test Failover in Hyper-V Replica creates test checkpoints – clean up after testing. Automate checkpoint management: Set-VM -CheckpointType Standard/Production/Disabled/ProductionOnly. Disable automatic checkpoints in production.

Q116 What is Virtual Fibre Channel in Hyper-V?

Answer : Hyper-V Virtual Fibre Channel (vFC) allows VMs to directly access SAN storage via Fibre Channel HBAs (Host Bus Adapters) on the host, appearing as dedicated FC ports to the VM. This enables: direct VM access to SAN volumes (not through host), live migration with FC storage (N_Port ID Virtualization – NPIV support on HBA), and running guest clustering over FC. Requires: Fibre Channel HBA with NPIV support, SAN zoning. Configure: Add-VMFibreChannelHba. Not as commonly used with modern iSCSI/SMB storage options.

Q117 What is Network Virtualization in Hyper-V?

Answer : Hyper-V Network Virtualization allows multiple VM networks with overlapping IP addresses to share the same physical network – similar to VLANs but more flexible. Uses NVGRE or VxLAN encapsulation to tunnel VM traffic over the physical network. Enables: multi-tenant isolation (cloud providers), moving VMs between subnets without reconfiguration, and flexible network topologies. Part of Microsoft SDN (Software Defined Networking). Managed by SCVMM network fabric or Windows Server Network Controller for complex deployments.

Q118 What are Hyper-V resource metering?

Answer : Hyper-V Resource Metering tracks VM resource usage for chargeback/showback in multi-tenant environments. Metrics tracked: average CPU usage, average/maximum memory pressure, outbound network traffic, minimum/average/maximum disk allocation. Enable: Enable-VMResourceMetering, collect data: Measure-VM, reset: Reset-VMResourceMetering. Data stored in Hyper-V WMI repository. Not persistent across host reboots – collect data regularly. Used with billing systems in hosting environments or for capacity planning.

Section 5: Storage & File Services

Questions 121 – 150

Q12 1 What is Storage Spaces in Windows Server?

Answer : Storage Spaces is a Windows Server feature that pools physical disks (HDDs, SSDs, NVMe) into a flexible storage pool, then creates virtual disks with resiliency. Resiliency types: Simple (no redundancy — striped), Mirror (2-way: 2 copies; 3-way: 3 copies), and Parity (distributed parity across 3+ disks). Supports: thin provisioning, tiering (SSD + HDD tiers), deduplication, and encryption. Managed via Storage Spaces control panel, Server Manager, or PowerShell (New-StoragePool, New-VirtualDisk). Suitable for small-scale deployments; use Storage Spaces Direct for scale-out.

Q12 2 What is Storage Spaces Direct (S2D)?

Answer : Storage Spaces Direct extends Storage Spaces to use directly attached storage across multiple cluster nodes — eliminating the need for a SAN or NAS. Each server contributes its local drives to a shared pool. S2D uses: SMB Direct (RDMA) for server-to-server communication, CSV (Cluster Shared Volumes) for shared namespace, and cache drives (NVMe or SSD) to accelerate spinning disk performance. Deploy with minimum 2 nodes (all-flash) or 3 nodes (hybrid), up to 16 nodes per cluster. Requires Datacenter edition. Basis for Azure Stack HCI.

Q12 3 What is the Scale-Out File Server (SOFS)?

Answer : Scale-Out File Server is a clustered file server role that provides continuously available file shares for application data, particularly Hyper-V over SMB and SQL Server over SMB. Unlike traditional file server clusters (one node active), SOFS uses active-active architecture — all cluster nodes simultaneously serve SMB shares. Uses CSV for shared storage. Benefits: linear scalability, automatic load balancing, no failover required (all nodes active), and transparent client reconnection. Requires at least 2 nodes, Datacenter edition recommended.

Q12 4 What is NTFS vs ReFS?

Answer : NTFS (NT File System): mature, feature-rich — supports ACLs, compression, EFS, quotas, hard links, symbolic links, sparse files, transaction support. Maximum volume: 256 TB. ReFS (Resilient File System): modern, designed for large storage — data integrity (checksums on all data and metadata), built-in RAID with Storage Spaces, integrity streams, block cloning (fast VM copy), sparse VDL (VM creation performance). ReFS limitations: no compression, no EFS, no quotas, not bootable. Use ReFS for: Hyper-V storage, Storage Spaces Direct, large file servers. Use NTFS for: OS volumes, applications requiring NTFS-specific features.

Q12
5

What is SMB (Server Message Block)?

Answer
:

SMB is the Windows file sharing protocol. Version history: SMBv1 (legacy, deprecated, insecure — disable it), SMBv2 (Windows Vista/2008 — major performance improvements, request compounding, larger MTU), SMBv3 (Windows 8/2012 — SMB Encryption, SMB Direct/RDMA, SMB Multichannel, transparent failover for SOFS). Check SMB version: `Get-SmbServerConfiguration`. Disable SMBv1: `Set-SmbServerConfiguration -EnableSMB1Protocol $false`. SMB uses TCP port 445 (modern) and 139 (older, NetBIOS). SMB over QUIC (Windows Server 2022) adds HTTPS-like SMB without VPN.

Q12
6

What is SMB Multichannel?

Answer
:

SMB Multichannel automatically uses multiple network connections (NICs) to increase SMB throughput and provide fault tolerance. Requirements: client and server both support SMB 3.0+, multiple network paths available (multiple NICs or RSS-capable NICs). With RDMA NICs: SMB Direct (zero-copy, kernel-bypass). SMB Multichannel is automatic — no configuration needed if requirements are met. Check: `Get-SmbMultichannelConnection`. Use for: Hyper-V over SMB (Scale-Out File Server), SQL Server over SMB, high-throughput file transfers.

Q12
7

What is DFS (Distributed File System)?

Answer
:

DFS provides a namespace for distributed file shares. Two components: DFS Namespaces (DFS-N) — creates a unified namespace (`\\domain\public\`) aggregating shares from multiple servers; clients see one path regardless of which server actually serves the data. DFS Replication (DFS-R) — replicates folders between servers for data availability and disaster recovery; uses compression (RDC — Remote Differential Compression); replaced FRS for SYSVOL replication. Together: high-availability file serving and data redundancy without complex failover clusters.

Q12
8

What is DFS Replication (DFS-R) and how does it work?

Answer
:

DFS-R is a multi-master replication engine using Remote Differential Compression (RDC) to replicate only changed blocks of files, minimizing bandwidth. Replication topology: Hub and Spoke (hub replicates to spokes), Full Mesh (each member replicates with all others), or custom. Staging area: temporary storage for files being replicated. Conflict resolution: last-writer-wins based on file modification time. Initial replication: uses pre-seeding for large datasets. Monitor: DFS Management console, `dfsrdiag.exe`, event logs (Event IDs 4612, 4104). Replaces FRS (File Replication Service) — never use FRS for new deployments.

Q12
9

What are NTFS permissions?

Answer : NTFS permissions control access to files and folders at the file system level. Basic permissions: Full Control, Modify, Read & Execute, List Folder Contents, Read, Write. Advanced (special) permissions: more granular control (Create Files, Delete Subfolders, Change Permissions, Take Ownership). Inheritance: permissions flow from parent folder to child objects by default – can be blocked. Effective permissions = cumulative of all NTFS permissions + share permissions (APPLY MORE RESTRICTIVE). Use `icacls` command or `Get-Acl/Set-Acl` PowerShell for management.

**Q13
0** What are share permissions vs NTFS permissions?

Answer : Share permissions apply when accessing resources over the network (not locally). Three levels: Read, Change, Full Control. NTFS permissions apply whether accessing locally or over the network. When accessing a shared folder over the network: effective access = MOST RESTRICTIVE of share permission AND NTFS permission. Best practice: Set share permission to Everyone: Full Control or Authenticated Users: Full Control, then control access with NTFS permissions only – this simplifies administration. For locally accessed shares, only NTFS permissions apply.

**Q13
1** What is iSCSI in Windows Server?

Answer : iSCSI (Internet Small Computer Systems Interface) provides block-level storage access over TCP/IP networks – essentially SCSI commands over Ethernet. Windows Server can act as: iSCSI Initiator (client – connects to iSCSI targets, assigns to server as local disks) via Microsoft iSCSI Initiator Service and `iscsicpl`. iSCSI Target Server (role – exports virtual disks as iSCSI targets to initiators). Use cases: providing shared storage to clusters without FC SAN, providing VM disk storage, backup targets. For performance: use dedicated network with jumbo frames (MTU 9000) and MPIO.

**Q13
2** What is MPIO (Multipath I/O)?

Answer : MPIO allows multiple physical paths between a server and storage device (SAN/NAS) for fault tolerance and load balancing. If one path fails, another takes over automatically. Load balancing policies: Round Robin, Round Robin with Subset, Least Queue Depth, Weighted Paths, Least Blocks. Windows Server: install MPIO feature, add MPIO support for specific devices (`Get-MSDSMSupportedHW`, `New-MSDSMSupportedHW`). DSM (Device Specific Module) may be required from storage vendor for non-standard devices. Required for iSCSI multipathing, FC SAN connections.

**Q13
3** What is Windows Server Deduplication?

Answer : Data Deduplication (dedup) identifies and eliminates duplicate data within a volume, storing only one copy of redundant chunks and replacing duplicates with references to the single copy. Use cases: file server volumes, Hyper-V VDI (virtual desktop infrastructure), software deployment servers. Not supported on: boot volumes, system volumes. Run-time: background optimization jobs run during off-peak hours. Savings: general file server 50-70%, VDI 80-95% savings typical. Enable: Enable-DedupVolume, check savings: Get-DedupStatus. Requires NTFS volumes.

Q13 4 What is Storage Tiering?

Answer : Storage Tiering in Storage Spaces automatically moves frequently accessed (hot) data to faster storage (SSD/NVMe) and less-accessed (cold) data to slower storage (HDD). Tiers are defined in the storage pool by media type. Tiering algorithm analyzes I/O patterns and optimizes placement during off-peak hours. Pin files/folders to specific tier: Set-FileStorageTier. Real-time tier (SSD write cache): absorbs write bursts before tiering optimization moves data. Combine with deduplication for maximum efficiency on large data sets.

Q13 5 What is Windows Server NFS?

Answer : Network File System (NFS) Server role in Windows Server provides file sharing for Linux/Unix clients using the NFS protocol. Windows Server supports NFS v3 and v4.1. Server for NFS: share Windows volumes with Linux/Unix systems. Client for NFS: mount NFS shares from Linux/NFS servers on Windows. Configure: Server Manager > File and Storage Services > Shares > New Share > NFS Share. Set Unix permission handling: Set-NfsMappedIdentity for user/group mapping. Useful for mixed Windows/Linux environments and Kubernetes persistent volume storage (NFS storage class).

Q13 6 What is the File Server Resource Manager (FSRM)?

Answer : FSRM is a Windows Server role for managing and classifying stored data. Features: Quota Management (limit disk space per folder with hard or soft limits, email notifications), File Screening (prevent users from saving specific file types – block .mp3, .exe), Storage Reports (generate scheduled or on-demand reports – duplicate files, largest files, files by type), File Classification Infrastructure (classify files with properties for downstream actions), and File Management Tasks (automated actions based on classification – archive, encrypt, expire files).

Q13 7 What are Offline Files and BranchCache?

Answer : Offline Files (Client-Side Caching): caches network share content locally, allowing access when disconnected. Configure via Group Policy. BranchCache: caches content from central servers at branch offices to reduce WAN bandwidth. Two modes: Distributed Cache (peers in same branch share cached content), Hosted Cache (dedicated cache server at branch). BranchCache supports: SMB, HTTP/HTTPS, BITS. Requires Windows Server 2008 R2+ for hosted cache server. Enable: BranchCache feature + Group Policy. Ideal for branch offices with limited bandwidth to central file servers.

Q13 8 What is Work Folders?

Answer : Work Folders is a Windows Server role that provides a sync solution for users to access corporate data from any device (PC, Mac, iOS, Android) — similar to consumer sync services but managed by IT. Data is stored on Windows file servers, synced to user devices. Features: data encryption on devices (EFS), selective wipe (wipe corporate data without affecting personal data), MDM integration, and access via HTTPS. Alternative to Dropbox/OneDrive for on-premises scenarios. Useful for users who need corporate data on personal devices without full VPN.

Q13 9 What is the new SMB over QUIC feature in Windows Server 2022?

Answer : SMB over QUIC provides a secure SMB tunnel over QUIC (Quick UDP Internet Connections) protocol using TLS 1.3, allowing users to access file shares over the internet without a VPN. Unlike traditional SMB over 445/TCP (often blocked by firewalls), SMB over QUIC uses UDP port 443 (HTTPS port). Requires: Windows Server 2022 Datacenter Azure Edition or Azure Stack HCI, certificate on file server, Windows 11 or patched Windows 10 clients. Features: end-to-end encryption, firewall-friendly (UDP 443), suitable for remote workers accessing corporate file shares securely.

Q14 0 What is the storage pool and virtual disk creation process?

Answer : Process: 1) Install disks in server. 2) Open Server Manager > File and Storage Services > Volumes > Storage Pools. 3) New Storage Pool — select physical disks. 4) New Virtual Disk from pool — choose resiliency (Simple/Mirror/Parity), size, provisioning (fixed/thin). 5) Initialize disk and create volume (partition/format). PowerShell: `New-StoragePool` □ `New-VirtualDisk` □ `Initialize-Disk` □ `New-Partition` □ `Format-Volume`. For Storage Spaces Direct: use `Enable-ClusterS2D` cmdlet which automates pool and tier creation.

Section 6: Network Policy Server & Remote Access

Questions 151 – 180

Q15
1 What is Network Policy Server (NPS)?

Answer : NPS is the Microsoft implementation of a RADIUS (Remote Authentication Dial-In User Service) server and proxy. It performs centralized authentication, authorization, and accounting for: VPN connections, wireless (802.1X), dial-up, and Remote Desktop Gateway connections. NPS evaluates connection requests against network policies (conditions + constraints + settings) to allow or deny access. NPS logs authentication attempts to Windows Event Log, SQL Server, or text files. NPS can proxy RADIUS requests to other RADIUS servers (NPS proxy).

Q15
2 What is a VPN and VPN protocols in Windows Server?

Answer : VPN (Virtual Private Network) creates encrypted tunnels over public networks. Windows Server VPN protocols: PPTP (old, weak encryption — avoid), L2TP/IPsec (strong, widely supported, requires certificate or pre-shared key), SSTP (SSL/TLS over 443 — firewall friendly), IKEv2 (modern, fast reconnect, VPN Reconnect feature for mobile users). DirectAccess is a VPN alternative. Configure via Routing and Remote Access Service (RRAS). Always-On VPN (AOVPN) is the modern replacement for DirectAccess using native VPN protocols.

Q15
3 What is DirectAccess?

Answer : DirectAccess provides seamless, always-on remote access to corporate network resources without a traditional VPN client — computers automatically connect when internet is available. Uses IPv6 (with ISATAP, 6to4, Teredo as transition technologies), IPsec, and computer certificates. Requirements: Windows 10/8.1 Enterprise or Windows 7 Ultimate/Enterprise clients, domain-joined, Windows Server 2012+ with DirectAccess role. Limitations: domain-joined clients only, no iOS/Android support, IPv6 dependency complexity. Being replaced by Always-On VPN.

Q15
4 What is Always-On VPN (AOVPN)?

Answer : Always-On VPN is Microsoft's modern successor to DirectAccess using standard VPN protocols (IKEv2, SSTP, L2TP). Features: user tunnel (user-authenticated, for domain resources), device tunnel (machine-authenticated, always connected — for pre-logon authentication, management traffic), Windows Autopilot support, Azure AD integration, conditional access, and support for non-domain-joined devices. Configure via Microsoft Intune, SCCM, or PowerShell deployment scripts. Supports Windows 10/11 clients. More flexible than DirectAccess.

Q15
5

What is 802.1X authentication for wireless?

Answer
:

802.1X is a port-based network access control standard for wired and wireless networks. Process: supplicant (client) connects to authenticator (switch/access point), which proxies authentication to an authentication server (NPS/RADIUS). EAP (Extensible Authentication Protocol) variants: PEAP-MSCHAPv2 (username/password, requires server certificate), EAP-TLS (certificate-based — most secure, requires client and server certificates), PEAP-TLS. Configure NPS as RADIUS server, wireless APs as RADIUS clients, and use Group Policy to configure client 802.1X profiles. Required for enterprise wireless security.

Q15
6

What are NPS Network Policies?

Answer
:

NPS Network Policies define conditions for granting or denying access. Each policy has: Conditions (day/time, client IP, user group, machine group, authentication type, called station ID), Constraints (idle timeout, session timeout, authentication methods allowed, IP filters), Settings (IP address assignment, routing, VLAN assignment via VSA — Vendor Specific Attributes). Policies evaluated top to bottom — first match applies. Default policies deny all access unless explicitly allowed. Use NPS for: VPN authorization, wireless access control, wired 802.1X NAC.

Q15
7

What is RADIUS proxy and NPS proxy?

Answer
:

NPS as a RADIUS proxy receives RADIUS authentication requests and forwards them to remote RADIUS servers. Use cases: centralized RADIUS server handling requests from multiple locations, forwarding requests to vendor-specific RADIUS (Cisco, Juniper), or load balancing across multiple NPS servers. Configure: NPS > RADIUS Proxy > Remote RADIUS Server Groups. Define routing rules based on user name (realm routing — username@realm forward to specific server group). NPS proxy doesn't evaluate policies — it only forwards to appropriate RADIUS server.

Q15
8

What is Routing and Remote Access Service (RRAS)?

Answer
:

RRAS is a Windows Server role providing routing and remote access capabilities. Features: VPN server (PPTP, L2TP, SSTP, IKEv2), Dial-up server (legacy), NAT (Network Address Translation — share internet connection), LAN routing (static routes, RIP, OSPF — via RRAS routing protocols), Site-to-Site VPN (connect two networks), and BGP support (Border Gateway Protocol — for cloud connectivity). Configure via RRAS MMC or PowerShell. For enterprise routing, dedicated network appliances are preferred; RRAS for simple NAT/VPN scenarios.

Q15
9

What is the Web Application Proxy (WAP)?

Answer : Web Application Proxy is a Windows Server role that provides reverse proxy capabilities for publishing internal web applications to external users. WAP works with AD FS to provide pre-authentication (users authenticate with corporate credentials before accessing application) or pass-through (authentication handled by published application). Published applications: HTTP/HTTPS, Microsoft Exchange, SharePoint, Lync/Skype, Office Web Apps. WAP replaces ISA Server/TMG for basic web publishing. Install on DMZ server separate from AD FS servers.

**Q16
0** What is Network Access Protection (NAP)?

Answer : NAP is a platform for enforcing health compliance on client computers before allowing access to the network. Health policy: antivirus up-to-date, Windows updates applied, firewall enabled. Non-compliant clients: placed in remediation network (access to remediation servers only). NAP enforcement methods: DHCP, VPN (RRAS), IPsec, 802.1X. Status: NAP was deprecated in Windows Server 2012 R2 and removed in Server 2016. Replaced by: Microsoft Intune compliance policies, Azure AD Conditional Access, Microsoft Defender for Endpoint. Not recommended for new deployments.

**Q16
1** What is Windows Server RD Gateway?

Answer : Remote Desktop Gateway (RD Gateway) enables secure remote access to internal RD Session Hosts, RemoteApp programs, and remote desktops from the internet using HTTPS (port 443). RD Gateway wraps RDP traffic in HTTPS, allowing access through firewalls that block RDP port 3389. Features: NPS integration for connection authorization policies (RD CAPs), resource authorization policies (RD RAPs — define which servers users can connect to), load balancing, and monitoring. Deploy in DMZ. Part of RDS infrastructure for external access.

**Q16
2** What is Network Load Balancing (NLB)?

Answer : NLB is a Windows Server feature that distributes incoming network traffic across multiple servers for scalability and high availability. NLB clusters: 2-32 nodes, all active simultaneously (active-active). Modes: Unicast (all nodes share same MAC — switch floods all nodes), Multicast (each node has unique MAC, NLB adds multicast MAC). Affinity modes: None, Single (same client always goes to same node), Network (same /24 subnet to same node). Use cases: web servers, VPN termination. Limitation: not suitable for database servers (use Failover Clustering instead). Being superseded by Azure Load Balancer and Application Gateway in hybrid/cloud.

**Q16
3** What is IPAM (IP Address Management)?

Answer : IPAM is a Windows Server role for managing and monitoring IP address space and DNS/DHCP infrastructure from a single console. Features: IP address inventory and utilization tracking, DNS zone management across multiple servers, DHCP server management across domain, IP address block management (subnets, IP ranges), automatic discovery of DHCP/DNS servers, and audit trail of IP assignments. IPAM does NOT replace DNS/DHCP servers – it provides centralized management. Requires SQL Server Express (included). Access control: role-based access with AD groups.

**Q16
4** What is the difference between split tunneling and full tunneling VPN?

Answer : Full Tunnel VPN: ALL client traffic (internet and corporate) routes through the VPN tunnel to corporate network. Benefits: full traffic inspection by corporate security, consistent filtering. Drawbacks: higher bandwidth on corporate internet connection, latency for internet-bound traffic. Split Tunnel VPN: only corporate-bound traffic routes through VPN; internet traffic goes directly from client. Benefits: lower bandwidth on corporate link, better performance for internet browsing. Drawbacks: corporate security controls don't apply to internet traffic. Always-On VPN supports both modes via routing configuration.

**Q16
5** What is IKEv2 VPN and why is it preferred?

Answer : IKEv2 (Internet Key Exchange version 2) is a modern, secure VPN protocol using IPsec. Advantages: VPN Reconnect (MOBIKE – seamlessly reconnects after brief network interruptions – essential for mobile users switching between WiFi and cellular), faster connection establishment, stronger encryption (AES-256, SHA-256, DH Group 14+), native support in Windows 10/11, iOS, macOS, Android. Disadvantages: may be blocked on networks blocking UDP 500/4500. Recommended for: Always-On VPN user tunnel, mobile devices. Use SSTP as fallback when IKEv2 is blocked.

**Q16
6** What is BGP in Windows Server RRAS?

Answer : Border Gateway Protocol (BGP) support in Windows Server RRAS allows dynamic routing between the Windows RRAS router and network infrastructure or cloud providers. Use cases: Azure Site-to-Site VPN with BGP (dynamic route advertisement), multi-site connectivity, and on-premises SDN deployments. Configure: Add-BgpRouter, Add-BgpPeer. BGP in Windows Server supports: eBGP (external BGP between autonomous systems), route policies, and integration with Azure VPN Gateway BGP peers. For complex enterprise BGP requirements, use dedicated hardware routers (Cisco, Juniper).

**Q16
7** What is SSTP VPN?

Answer : Secure Socket Tunneling Protocol (SSTP) creates a VPN tunnel using HTTPS (SSL/TLS) over TCP port 443. Since port 443 is rarely blocked by firewalls (it's the standard HTTPS port), SSTP works in environments where PPTP, L2TP, and IKEv2 are blocked. Requires: server SSL certificate, Windows Vista SP1+ clients. Limitations: Windows-only native support (no native iOS/macOS SSTP), slightly higher overhead than IKEv2, no support for VPN Reconnect. Use SSTP as fallback for Always-On VPN when IKEv2 is unavailable. Not recommended as primary protocol for mobile users.

Q16₈ What is RD Session Host (RDSH) and RD Connection Broker?

Answer : RD Session Host: Windows Server running session-based Remote Desktop — multiple users connect simultaneously, each getting their own Windows session on the shared server. RD Connection Broker: manages user sessions for RDS deployments — routes new connections to appropriate RDSH (load balancing), reconnects users to existing sessions (session persistence), and provides single-instance connection point. In high-availability deployments, Connection Broker uses SQL Server for session state and is load balanced. Required for enterprise RDS deployments with multiple RDSH servers.

Q16₉ What is the RD Licensing role?

Answer : RD Licensing manages Remote Desktop Services Client Access Licenses (RDS CALs). Each user or device connecting to RD Session Host, RD Virtualization Host requires an RDS CAL. Two CAL types: RDS Per User CAL (user-based, roaming), RDS Per Device CAL (device-based, all users from that device). Grace period: 120 days without licensing server before connections blocked. Configure via Group Policy: specify licensing server address and mode. RD Licensing server must be accessible by RD Session Hosts. Install via Server Manager > Remote Desktop Services > RD Licensing.

Q17₀ What is SD-WAN integration with Windows Server?

Answer : Software-Defined WAN extends RRAS and networking capabilities to optimize connectivity between branch offices and cloud. Windows Server RRAS with BGP can peer with SD-WAN appliances. Azure Virtual WAN and RRAS integration: use S2S VPN from RRAS to Azure Virtual Hub. Windows Server NLB and RRAS ECMP (Equal Cost Multi-Path) distribute traffic across multiple WAN links. For enterprise SD-WAN, specialized solutions (Cisco Viptela, VMware SD-WAN, Palo Alto Prisma) are preferred. Windows Server provides a cost-effective SD-WAN endpoint for smaller branch offices.

Section 7: Windows Server Security

Questions 181 – 210

Q18
1 What is Windows Server security hardening?

Answer : Security hardening reduces the attack surface of Windows Server. Key practices: Disable unused services and roles, apply CIS Benchmarks or STIG configurations, enable Windows Defender Firewall on all profiles, enable Windows Defender Antivirus, use Server Core (no GUI), disable SMBv1, enforce strong password policies, enable audit policies, require MFA for admin accounts, apply security baselines via GPO (Microsoft Security Compliance Toolkit), implement LAPS (Local Administrator Password Solution), and keep servers patched with current updates.

Q18
2 What is Microsoft Security Compliance Toolkit (SCT)?

Answer : SCT provides security configuration baselines for Windows Server, Windows client, Microsoft 365 Apps, and other products. Baselines are GPO templates representing Microsoft's recommended security configurations. Tools included: Policy Analyzer (compare GPOs, identify differences), Local Group Policy Object (LGPO) utility (import/export local policies), Set Object Security utility. Download from Microsoft Download Center. Apply baselines as starting point and customize for organizational requirements. Baselines address common vulnerabilities and align with CIS benchmarks and STIG.

Q18
3 What is LAPS (Local Administrator Password Solution)?

Answer : LAPS manages local administrator account passwords for domain-joined computers — generating random, unique passwords for each machine and storing them in Active Directory (dsAttr). This eliminates the security risk of all machines using the same local admin password (pass-the-hash lateral movement). Windows LAPS (built-in since Windows Server 2022/Windows 11 22H2): stores passwords in Azure AD or AD DS, supports password history, encrypted password storage. Legacy LAPS: requires schema extension and separate client agent. Configure via Group Policy.

Q18
4 What is Credential Guard?

Answer : Windows Defender Credential Guard uses Virtualization-Based Security (VBS) to isolate LSA (Local Security Authority) secrets in a virtualized container (Virtual Secure Mode), protecting against pass-the-hash and pass-the-ticket attacks. Isolated LSA process stores: NTLM password hashes, Kerberos TGTs, domain credentials. Even if kernel is compromised, credentials in VSM cannot be accessed. Requirements: UEFI 2.3.1+, Secure Boot, 64-bit CPU with virtualization extensions, IOMMU. Enable via Group Policy: Computer Configuration > Administrative Templates > System > Device Guard.

Q18
5

What is Device Guard / Windows Defender Application Control?

Answer
:

Windows Defender Application Control (WDAC, formerly Device Guard) restricts which applications can run on a device – allowing only trusted, signed applications. Uses code integrity policies to allow only approved binaries. WDAC is superior to AppLocker: kernel-level enforcement, can't be bypassed by local admins. Deploy policies via Group Policy (in audit mode first, then enforce). WDAC uses Virtualization-Based Code Integrity (HVCI) when available. Consider WDAC for: servers running only specific applications, VDI environments, high-security workstations.

Q18
6

What is Privileged Access Workstation (PAW)?

Answer
:

A PAW is a dedicated workstation used exclusively for administrative tasks – never for browsing internet, email, or general use. PAW principles: clean source (admin credentials only enter PAW, never shared computers), separation (admin-only workstation with hardened configuration, no internet, email). Benefits: isolates admin credentials from credential theft risks on general-purpose computers. Implementation: separate physical hardware or Credential Guard + WDAC hardened VM. Required as part of Microsoft's Privileged Access strategy for tier-0 (Domain Admins, enterprise admins) administrative work.

Q18
7

What is Tiered Administration Model?

Answer
:

Microsoft's Tiered Administration Model separates privileged accounts into tiers to prevent credential theft from compromising higher-tier assets: Tier 0 – Domain Controllers, AD DS, AD FS, Azure AD Connect (most critical). Tier 1 – Servers and applications. Tier 2 – User workstations and devices. Rules: Tier 0 admins only administer Tier 0 systems (never log into workstations), Tier 1 admins only administer Tier 1 systems, Tier 2 admins handle workstations. Prevent credential exposure: use Admin Workstations per tier, never use Domain Admin for workstation management. Enforced via GPO (Deny log on locally/via RDP at lower tiers).

Q18
8

What is Windows Event Auditing?

Answer
:

Windows Server auditing records security-relevant events in the Security Event Log. Configure via Group Policy: Advanced Audit Policy Configuration (more granular than Basic Audit). Key categories: Logon/Logoff (4624 success, 4625 failure), Account Logon (Kerberos authentication), Account Management (4720 user created, 4740 lockout), Object Access (file/folder access – requires SACL on resource), Process Creation (4688 – logs command-line arguments when enabled), Privilege Use, Policy Change, System. Critical to monitor: 4625 (failed logons), 4768/4771 (Kerberos), 4776 (NTLM).

Q18
9

What is the Security Configuration Wizard (SCW)?

Answer : SCW is a Windows Server tool that creates security policies based on the server's actual role – disabling unused services, blocking unused ports, and restricting IIS settings. It analyzes installed roles and features and generates a minimally required security policy. Policies saved as XML and applied via command line (scwcmd). While SCW is still available, Microsoft now recommends: Server Core installation (eliminates GUI attack surface), Security Compliance Toolkit baselines, and DISM for component removal. SCW is particularly useful for hardening servers with specific role combinations.

Q190 What is IPsec in Windows Server?

Answer : IPsec (Internet Protocol Security) provides authentication and encryption at the network layer (IP level). Windows Server uses IPsec for: secure communication between servers (server isolation), domain isolation (only domain members can communicate), protecting sensitive data in transit, and VPN (L2TP/IPsec, IKEv2). Configure via Windows Defender Firewall Connection Security Rules or Group Policy (IP Security Policies). Modes: Transport (encrypts payload, not header), Tunnel (encrypts entire packet – used for VPN). Use IPsec for: server-to-server encryption, isolating DCs from non-compliant hosts.

Q191 What is Windows Defender ATP / Microsoft Defender for Endpoint?

Answer : Microsoft Defender for Endpoint (MDE) is an enterprise endpoint security platform providing: threat and vulnerability management, attack surface reduction, next-gen antivirus (Windows Defender), endpoint detection and response (EDR), automated investigation and remediation, and Microsoft Threat Intelligence integration. Onboard servers via: SCCM, Group Policy, Intune, or local script. MDE provides visibility into: process trees, network connections, file operations, registry changes, and user logon activities. Key for detecting post-exploitation behavior that signature-based AV misses.

Q192 What is RBAC for Windows Server administration?

Answer : Role-Based Access Control for server administration uses security groups and access control lists (ACLs) to grant minimum required permissions. Key approaches: Just Enough Administration (JEA) – constrains which PowerShell commands a user can run via constrained runspace. Delegation of Control – grant specific AD administrative rights (reset passwords, manage specific OUs) without full admin. Local Admin groups – use domain security groups for local admin access (no generic 'administrator' account). LAPS – unique local admin passwords. Privileged Identity Management – just-in-time admin access via PIM.

Q193 What is Just Enough Administration (JEA)?

Answer : JEA is a PowerShell security feature that allows non-privileged users to run specific PowerShell commands as a privileged virtual account, without granting full administrative rights. Configuration: Role Capability Files (.psrc) define allowed commands, Session Configuration Files (.pssc) define who can use which roles. Users connect to JEA endpoint via PSSession. All actions logged to transcript files. Example: helpdesk staff can reset AD passwords and restart specific services without Domain Admin rights. Deploy via DSC or Group Policy. Critical for tier-based administration.

Q19 4 What is Windows Server Protected Users group behavior?

Answer : Protected Users security group (requires 2012 R2+ DCs and 2012 R2+ clients) provides automatic protections: No NTLM authentication (Kerberos only), no DES or RC4 Kerberos encryption (AES 128/256 only), no CredSSP credential delegation, no WDigest authentication (no plaintext caching), Kerberos TGT lifetime capped at 4 hours (non-renewable), no credential caching in LSASS. Add to Protected Users: Domain Admins and other privileged accounts. Test carefully — applications requiring NTLM, delegation, or long-lived Kerberos tickets will break.

Q19 5 What is the Security Account Manager (SAM) database?

Answer : The SAM database stores local user account information and password hashes for standalone computers and member servers (not domain accounts — those are in NTDS.DIT on DCs). Located at: %SystemRoot%\System32\Config\SAM (locked during Windows operation). Password hashes: NTLM hash (32-byte MD4 of Unicode password). The SAM is protected by: file system ACL, registry key encryption (SYSKEY/LSA secrets), and the fact that the file is locked. Remote access to SAM is blocked by default (RestrictRemoteSAM Group Policy). Attack: extract with tools like mimikatz (requires admin/SYSTEM).

Q19 6 What is Microsoft Baseline Security Analyzer (MBSA)?

Answer : MBSA (Microsoft Baseline Security Analyzer) is a deprecated tool that scanned Windows systems for missing security updates and common security misconfigurations. MBSA has been discontinued and replaced by: Microsoft Defender Vulnerability Management (part of MDE), Microsoft Secure Score, Security Center in Microsoft Defender for Cloud (Azure), and Windows Update Compliance (via Azure Monitor). For patch compliance scanning: use Windows Admin Center, Defender Vulnerability Management, or Qualys/Tenable for comprehensive vulnerability assessment.

Q19 7 What is Secure Admin Workstation (SAW) vs PAW?

Answer : PAW (Privileged Access Workstation): dedicated physical workstation for administration only — no internet, email, general use. Highest security, most friction for admins. SAW (Secure Admin Workstation): general-purpose workstation with enhanced security controls (application whitelisting, enhanced logging, restricted internet access) used for both admin and regular work — compromise between security and usability. Microsoft recommends PAW for Tier 0/1 admin tasks. For organizations unable to provide dedicated physical PAWs, a secured admin VM with Credential Guard on a standard workstation is an acceptable intermediate.

Q19
8 What is Windows Defender Exploit Guard?

Answer : Windows Defender Exploit Guard provides four components for attack surface reduction: Attack Surface Reduction (ASR) rules — block malicious behaviors (Office macros downloading executables, credential theft from LSASS, ransomware-like behavior); Exploit Protection (replaces EMET) — per-process mitigations (DEP, ASLR, CFG, stack pivot protection); Network Protection — block connections to known malicious domains (based on Windows Defender SmartScreen intelligence); Controlled Folder Access — protect specific folders from unauthorized write access (ransomware protection). Configure via Group Policy, Intune, or PowerShell.

Q19
9 What is AppLocker?

Answer : AppLocker provides application control policies that specify which applications users can run. Rule collections: Executable files (.exe, .com), Windows Installer (.msi, .msp), Scripts (.ps1, .bat, .vbs, .js), Packaged apps (UWP Store apps), DLL files. Rule conditions: publisher (signed code), path, file hash. Can be set to: Allow (whitelist — default deny), Deny (blacklist). Audit mode: log violations without blocking. Policy applied via Group Policy. Limitations: can be bypassed by local admins. For stronger enforcement, use Windows Defender Application Control (WDAC). AppLocker is suitable for standard user environments.

Q20
0 What are common Windows Server security event IDs to monitor?

Answer : Critical event IDs: 4624 (successful logon — note LogonType 3=network, 10=remote interactive), 4625 (failed logon — investigate multiple failures), 4648 (logon with explicit credentials — runas, pass-the-hash), 4672 (special privileges assigned — admin logon), 4740 (account locked out), 4756/4757 (universal group membership change), 4768/4769 (Kerberos TGT/ST requests), 4776 (NTLM credential validation), 4799 (local group enumeration — recon activity), 7045 (new service installed), 4697 (service installed — creation of persistence), 1102 (audit log cleared — major red flag).

Section 8: Windows Server High Availability

Questions 211 – 240

Q21
1

What is Windows Server Failover Clustering (WSFC)?

Answer
:

Windows Server Failover Clustering provides high availability for services and applications by grouping multiple servers into a cluster that automatically fails over workloads when a node fails. Architecture: cluster nodes share storage (SAN, CSV, or S2D), maintain quorum (voting to determine cluster health), and use cluster networks for heartbeat and communication. Common use cases: SQL Server Always On FCI, Scale-Out File Server, Hyper-V VM HA, Exchange DAG, SAP clusters. Requires: same hardware platform, same Windows Server version, domain-joined servers.

Q21
2

What is quorum in a failover cluster?

Answer
:

Quorum determines when a cluster has enough healthy members to remain operational. Without quorum, cluster stops to prevent split-brain (two isolated parts both thinking they're the cluster). Quorum modes: Node Majority (odd number of nodes), Node and Disk Witness (even nodes + shared disk vote), Node and File Share Witness (even nodes + SMB share on remote server), No Majority: Disk Only (single shared disk owns quorum — not recommended). Dynamic Quorum: automatically adjusts votes as nodes leave/join. Best practice: configure cloud witness (Azure Blob Storage) for clusters spanning datacenters.

Q21
3

What is a cloud witness in failover clustering?

Answer
:

Cloud Witness uses Azure Blob Storage as a quorum witness for Windows Server Failover Clusters — replacing disk witness and file share witness for geographically dispersed clusters. Benefits: no single point of failure (Azure handles availability), cost-effective (uses Azure Storage Account), works across datacenters and cloud, no additional server required. Requirements: Windows Server 2016+, Azure subscription, public internet connectivity or ExpressRoute/VPN to Azure Storage. Configure in Cluster Manager or Set-ClusterQuorum -CloudWitness. Uses standard Azure Blob Storage (not premium).

Q21
4

What is Active/Active vs Active/Passive clustering?

Answer : Active/Passive: one node actively runs the workload; standby node activates only during failover. Simple, predictable resource allocation, but standby node resources wasted during normal operation. Active/Active: all nodes simultaneously run workloads – different cluster groups on different nodes. Better resource utilization, but capacity planning must account for 'N-1' (remaining nodes handle all workload when one fails). WSFC supports both: most SQL Server FCI deployments are active/passive; Hyper-V clusters can be active/active (VMs distributed across nodes).

**Q21
5** What are cluster roles and cluster resources?

Answer : Cluster resources are individual components managed by the cluster (IP address, network name, disk, application). Cluster roles (formerly cluster applications/services) are groups of related resources that fail over together. Built-in cluster roles: Generic Service, Generic Application, Generic Script, File Server (Standard and SOFS), Hyper-V, iSCSI Target, VM, DTC. Role properties: preferred owner (which node should run it), possible owners (which nodes can run it), failover settings (how quickly to fail over, how many times). Move role: Move-ClusterGroup PowerShell or drag in Cluster Manager.

**Q21
6** What is SQL Server Always On Failover Cluster Instance (FCI)?

Answer : SQL Server FCI is a single SQL Server instance that spans multiple cluster nodes. The SQL instance uses shared storage (SAN, CSV) and fails over between nodes along with the shared disk. One node active at a time (active/passive). SQL Server runs on the virtual cluster name/IP. Failover: automatic (if heartbeat lost) or manual. Difference from Always On AG: FCI is single shared-storage instance (failover but no readable secondary), AG is separate instances with database-level replication and readable secondaries. FCIs provide instance-level HA; AGs provide database-level HA with read scale.

**Q21
7** What is SQL Server Always On Availability Groups?

Answer : Always On Availability Groups (AG) is a SQL Server HA solution providing database-level protection without shared storage. Primary replica: reads and writes. Secondary replicas: receive log-stream replication, can be readable (for reporting), up to 8 replicas (SQL Server 2019+). Failover modes: Automatic (synchronous secondary – no data loss), Manual (asynchronous – potential data loss). Listener: virtual name/IP clients connect to – routes to primary or readable secondary. Requires: WSFC membership (or Basic AG without WSFC in SQL 2016+), SQL Server Enterprise edition (Basic AG in Standard). AG spans datacenter for DR.

**Q21
8** What is Network Load Balancing (NLB) vs Failover Clustering?

Answer : NLB: active-active, all nodes serve traffic simultaneously, suitable for stateless workloads (web, VPN), no shared storage, simple setup. Failover Clustering: active-passive or active-active with shared storage, designed for stateful applications (SQL, file server, Hyper-V), provides automatic failover with minimal downtime, requires shared storage (except Hyper-V S2D). Use NLB for: web servers, DNS servers, VPN termination — where requests are stateless and load distribution is needed. Use Failover Clustering for: databases, file servers, VMs — where state and storage must be protected.

Q21
9 What is stretched cluster (geo-cluster)?

Answer : A stretched cluster spans two or more physical sites for disaster recovery. Nodes in Site A and Site B are in the same cluster, storage replicated between sites using synchronous (for near-zero RPO) or asynchronous replication. Options: Storage Replica (built-in Windows Server Datacenter) with Failover Clustering, SAN-based replication (array replication), or S2D stretched cluster. For automatic failover between sites: synchronous replication + majority quorum with file share witness in 3rd site. Challenge: split-brain prevention when WAN link fails. Azure Site Recovery provides cloud-based stretched cluster alternative.

Q22
0 What is Storage Replica?

Answer : Storage Replica is a Windows Server Datacenter feature providing synchronous or asynchronous block-level replication between servers or clusters. Synchronous: I/O acknowledged only after committed to both source and destination — zero data loss, adds write latency. Asynchronous: I/O acknowledged after source commit, replication follows — minimal latency impact, potential data loss. Use cases: stretch clusters, server-to-server DR, cluster-to-cluster replication. Configure: New-SRPartnership PowerShell cmdlet. Destination volume is offline (read-only) during replication. Requires at least two dedicated log volumes (one per server).

Q22
1 What is the difference between RTO and RPO?

Answer : RTO (Recovery Time Objective): maximum acceptable time to restore service after a disaster — 'how long can we be down?' Lower RTO requires faster recovery mechanisms (hot standby, automatic failover). RPO (Recovery Point Objective): maximum acceptable data loss measured in time — 'how much data can we lose?' Lower RPO requires more frequent backups or synchronous replication. Example: RTO=4 hours, RPO=1 hour means: system restored within 4 hours, at most 1 hour of data lost. Synchronous replication achieves RPO=0. WSFC with SQL AG (synchronous) can achieve RTO<1 minute, RPO=0.

Q22
2 What is Azure Site Recovery (ASR) for Windows Server?

Answer : Azure Site Recovery provides disaster recovery by replicating on-premises physical servers, VMware VMs, and Hyper-V VMs to Azure. Process: install Mobility Service on servers, configure replication policy, ASR replicates to Azure Storage. Failover: RTO typically 15-30 minutes. Test failover: non-disruptive DR drill in isolated Azure network. Failback: return to on-premises after primary site restored. ASR handles: RPO as low as 30 seconds for VMware, recovery point history, app-consistent snapshots, network mapping. Cost: ASR license per protected server/VM plus Azure storage for replica.

**Q22
3** What is Windows Server Azure Extended Network?

Answer : Azure Extended Network allows on-premises workloads to be migrated to Azure while retaining their original IP addresses – no reconfiguration of applications, firewall rules, or clients. Uses: IP-HTTPS tunnel between on-premises network and Azure VNet, virtual NIC in Azure VM appears on the same subnet as on-premises. Component: Azure Network Adapter extension in Windows Admin Center. Use cases: lift-and-shift migration without IP address changes, gradual migration with some VMs in Azure keeping on-prem IPs. Requires: Azure subscription, VPN or ExpressRoute, Windows Admin Center.

**Q22
4** What is the Windows Server migration tool?

Answer : Windows Server Migration Tools (WSMT) help migrate specific server roles and features from older Windows Server versions to newer ones. Migrating: DHCP server configuration, DNS server, Local Users and Groups, IP configuration, NLB, Print and Document Services, Web Server (IIS), and FileServices data. Process: install WSMT on source and destination, export role data to migration store, import on destination. For full server migrations: Azure Migrate, VMware HCX, or Storage Migration Service (SMS). SMS migrates file server data while remapping access paths transparently.

**Q22
5** What is Storage Migration Service?

Answer : Storage Migration Service (SMS) is a Windows Server 2019+ feature that migrates file servers to newer hardware without users needing to know the migration occurred. SMS: inventories source servers, copies data (including NTFS permissions, shares, local users, local groups), sets up a proxy that redirects client connections from old server to new server (cutover), then performs a final sync and cutover. Supports migrating from: Windows Server 2003+, NetApp ONTAP, EMC Isilon. Managed via Windows Admin Center or PowerShell. Enables hardware refreshes with minimal downtime.

**Q22
6** What is planned vs unplanned failover?

Answer : Planned failover: administrator-initiated, performed for maintenance (patching, hardware replacement), no data loss expected, services briefly unavailable during switchover, clean shutdown of active role before activation on standby. Examples: Move-ClusterGroup, SQL AG manual failover, Hyper-V planned failover. Unplanned failover: occurs automatically when active node fails unexpectedly, potential brief data loss depending on transaction logs, cluster detects node failure via heartbeat timeout, cluster service automatically starts role on another node. Monitor: Cluster Events in Event Viewer (1069 – resource failure, 1135 – node removed from cluster).

Q22
7 What is the cluster validation wizard?

Answer : The Cluster Validation Wizard (Validate-Cluster PowerShell, Test-Cluster, or Cluster Manager > Validate) runs a comprehensive set of tests to verify cluster hardware and software configuration before or after creating a cluster. Test categories: Inventory, Network, Storage, System Configuration, Hyper-V (if applicable). Microsoft only supports failover clusters that pass all validation tests (supported configuration requirement). Run validation: after hardware changes, before and after software updates, and when troubleshooting cluster issues. Use -Include or -Exclude to run specific test categories.

Q22
8 What is graceful node drain and cluster-aware updating (CAU)?

Answer : Graceful node drain: before taking a node offline for maintenance, live migrate all VMs to other nodes (Suspend-ClusterNode -Drain). CAU (Cluster-Aware Updating): automates applying OS updates to cluster nodes without manual intervention. CAU orchestrates: draining node, installing updates, rebooting, resuming node, moving to next node – maintaining application availability throughout patching. CAU can run in self-updating mode (cluster updates itself) or remote-updating mode (external system triggers updates). Integrates with WSUS/Windows Update. Essential for maintaining cluster uptime during regular patching cycles.

Q22
9 What are cluster network roles?

Answer : Cluster networks are assigned roles that determine how they're used: Cluster Only (heartbeat/internal cluster communication – isolated from client traffic), Client and Cluster (used for both application access and internal communication – not recommended for production), Client Only (not used for cluster communication). Best practice: dedicated NIC(s) for cluster heartbeat (Cluster Only) separate from application traffic (Client and Cluster). Multiple cluster heartbeat networks increase fault tolerance (cluster fails over only if ALL heartbeat networks fail). Configure in Failover Cluster Manager > Networks.

Q23
0 What is PowerShell Failover Clustering commands?

Answer : Key PowerShell commands: Get-Cluster (cluster info), Get-ClusterNode (node status), Get-ClusterGroup (cluster roles/apps), Get-ClusterResource (individual resources), Move-ClusterGroup (manually move role to another node), Suspend-ClusterNode -Drain (drain VMs before maintenance), Resume-ClusterNode (return node to service), Start-ClusterGroup / Stop-ClusterGroup (start/stop roles), New-Cluster (create cluster), Add-ClusterNode (add node), Test-Cluster (run validation), Set-ClusterQuorum (configure quorum), Get-ClusterLog (collect cluster log). Always use PowerShell for scripted cluster operations.

Section 9: Windows Server Monitoring & Troubleshooting

Questions 241 – 270

**Q24
1** What performance monitoring tools are available in Windows Server?

Answer : Performance monitoring tools: Performance Monitor (perfmon.exe) – built-in tool for real-time and historical metric collection, graphing, and logging. Key counters: Processor\% Processor Time, Memory\Available MBytes, PhysicalDisk\Avg. Disk Queue Length, Network Interface\Bytes Total/sec. Data Collector Sets: automated collection of performance counters to files. Resource Monitor (resmon.exe) – real-time view of CPU, memory, disk, and network by process. Task Manager – quick process view. Windows Admin Center – modern web-based performance dashboard with historical charts.

**Q24
2** What is the Windows Reliability Monitor?

Answer : Reliability Monitor (perfmon /rel) shows a stability index (1-10) graph over time and lists events that caused reliability changes: application failures, Windows failures, hardware failures, warnings, information. Reliability index decreases on failures and gradually increases when system is stable. Useful for: identifying patterns of instability, correlating problems with software installations or updates, and communicating system health to users or management. Access via Control Panel > Security and Maintenance > Reliability History or from Resource Monitor.

**Q24
3** How do you use Process Monitor (ProcMon)?

Answer : Process Monitor (SysInternals – sysinternals.com) captures real-time file system, registry, process, network, and profiling activity on Windows. Essential for troubleshooting: application installation failures (missing DLLs, permission denied), application crashes (identify what file/registry access fails), performance issues (identify slow operations), and security incidents (identify what files/registry keys malware accesses). Filters are essential – ProcMon captures everything, filter by process name, path, result (ACCESS DENIED). Save traces for offline analysis. Combine with Process Explorer for comprehensive process investigation.

**Q24
4** What is the Windows Memory Diagnostic tool?

Answer : Windows Memory Diagnostic (mdsched.exe) scans RAM for hardware errors. Tests run during system restart, before Windows loads. Results: displayed in Event Viewer > Windows Logs > System (Event Source: MemoryDiagnostics-Results, Event ID 1201 for results). Extended test: more comprehensive but takes longer. Use when: system crashes (BSODs), application crashes with out-of-memory errors, or random system instability. For more comprehensive testing: MemTest86+ (bootable, multiple passes, tests all RAM). RAM failures are often intermittent – run multiple passes.

Q24
5 What is BlueScreen (BSOD) and how do you analyze it?

Answer : Blue Screen of Death (BSOD) is a critical system error (Stop error) that appears when Windows detects an unrecoverable error. BSOD contains: stop code (e.g., IRQL_NOT_LESS_OR_EQUAL, PAGE_FAULT_IN_NONPAGED_AREA), parameters, and crashed module. Analysis: configure system to create memory dumps (Control Panel > System > Advanced > Startup and Recovery – set to Kernel Memory Dump or Complete Memory Dump). Analyze dump with WinDbg (Windows Debugger), !analyze -v command. Common causes: driver bugs (most common), hardware failures (RAM, disk), overclocking, software bugs. Use WinDbg Preview from Microsoft Store.

Q24
6 What is Event Viewer and how do you use it for troubleshooting?

Answer : Event Viewer (eventvwr.msc) provides access to Windows event logs. Key logs: System (service failures, driver issues, hardware errors), Application (application events, crashes), Security (audit events), Setup. Applications and Services Logs contain role/application-specific logs (AD DS, DNS, DHCP, IIS, Hyper-V). Filtering: right-click log > Filter Current Log – filter by level (Error, Warning), event source, event ID, time range. Creating custom views: save filtered views for quick access. Event subscriptions: collect events from remote computers to central server. Use PowerShell Get-WinEvent for scripted log analysis.

Q24
7 What is WMI and how do you troubleshoot WMI issues?

Answer : WMI issues manifest as: Event Viewer errors (Source: WinMgmt), monitoring tool failures, SMS/SCCM collection failures, slow/hanging WMI queries. Diagnosis: winmgmt /verifyrepository (check repository integrity), winmgmt /salvagerepository (repair inconsistencies), winmgmt /resetrepository (reset – last resort, causes data loss). Check WMI service: Get-Service winmgmt. High CPU from WMI Provider Host (WmiPrvSE.exe): identify provider using wbemtest or WMI Explorer. Restart: net stop winmgmt, net start winmgmt. Advanced: wmidia.vbs tool for comprehensive WMI diagnostics.

Q24
8 What are Windows Server network troubleshooting commands?

Answer : Essential network troubleshooting: ipconfig /all (IP config, DNS, DHCP), ping (connectivity), tracert (path), pathping (path + packet loss), nslookup/Resolve-DnsName (DNS), netstat -an / ss (connections and ports), arp -a (ARP cache), route print (routing table), netsh interface show interface (NIC status), Test-NetConnection host -Port 443 (PowerShell — tests TCP connectivity), Get-NetAdapter (adapter status), Get-NetIPConfiguration, telnet host port (test port — install first), netsh wlan show all (wireless). For WAN links: tracert, mtr (Matt's traceroute), ping with -t flag.

Q24
9 What is SFC and DISM?

Answer : System File Checker (SFC): scans and repairs corrupted Windows system files. Run: sfc /scannow (requires admin, may need offline mode for boot files). Results in CBS.log (%windir%\Logs\CBS\CBS.log). DISM (Deployment Image Servicing and Management): repairs the Windows component store (WinSxS) which SFC uses as source. Run first: DISM /Online /Cleanup-Image /ScanHealth (scan), DISM /Online /Cleanup-Image /CheckHealth, DISM /Online /Cleanup-Image /RestoreHealth. Best practice: run DISM RestoreHealth first (repairs component store), then run SFC. Together they fix most Windows system file corruption.

Q25
0 What is the netsh command?

Answer : Netsh (Network Shell) is a command-line scripting utility for configuring and displaying network settings. Common uses: netsh int ip reset (reset TCP/IP stack), netsh winsock reset (reset Winsock catalog — fixes corrupt TCP/IP), netsh advfirewall set allprofiles state on/off (firewall control), netsh interface show interface (NIC status), netsh http show urlacl (URL reservations), netsh trace start capture=yes (packet capture), netsh branchcache show status. Netsh has been largely replaced by PowerShell network cmdlets but remains useful for specific scenarios and legacy compatibility.

Q25
1 What is Windows Performance Recorder (WPR) and Analyzer (WPA)?

Answer : WPR (Windows Performance Recorder) captures detailed ETW (Event Tracing for Windows) traces of system activity — CPU, disk, memory, network at very granular level. WPA (Windows Performance Analyzer) visualizes and analyzes WPR traces. Part of the Windows Assessment and Deployment Kit (ADK). Use for: identifying slow boot causes, application performance issues, high CPU attribution at the kernel/driver level, memory leak investigation. Better than perfmon for identifying root causes. Microsoft uses these tools internally. Available in Windows SDK/ADK download. Generates large trace files (hundreds of MB per minute).

Q25
2 What are Windows Server startup modes for troubleshooting?

Answer : Startup modes: Safe Mode — loads minimal drivers and services (no network). Safe Mode with Networking — Safe Mode plus network drivers. Safe Mode with Command Prompt — Safe Mode with cmd.exe instead of Explorer. Last Known Good Configuration — loads last working registry configuration (limited usefulness in modern Windows). Recovery Mode — Windows RE (Recovery Environment): system restore, startup repair, command prompt, system image recovery, refresh/reset. DSRM (Directory Services Restore Mode) — for AD DS troubleshooting on DCs. Access recovery options: F8 during boot (less reliable in fast boot), shift+restart from Windows, bcdedit /set {default} safeboot minimal.

Q25
3 What is System Center Operations Manager (SCOM)?

Answer : SCOM is an enterprise monitoring solution for Windows Server and other systems. Features: proactive monitoring (management packs define what to monitor — thresholds, alerts), performance data collection, distributed application monitoring, alert management workflow, dashboards and reports, network device monitoring, Unix/Linux monitoring. Management Packs available for: Windows Server roles, SQL Server, Exchange, AD DS, IIS, Hyper-V, Azure. Requires: SQL Server (for Operations and Data Warehouse databases), Management Server, Gateway Server (for untrusted networks). SCOM is being superseded by Azure Monitor and Microsoft Sentinel in cloud-first organizations.

Q25
4 What is the Windows System Assessment Tool (WinSAT)?

Answer : WinSAT evaluates Windows performance capabilities and generates a Windows Experience Index (WEI) score (1.0–9.9) across: processor, memory, graphics, gaming graphics, and primary hard disk. Run: winsat formal (full assessment, takes several minutes). Results: Get-CimInstance Win32_WinSAT. Useful for: hardware baseline documentation, identifying performance bottlenecks, comparing hardware capability before/after upgrades. Note: WEI score is no longer displayed in Windows 10/11 GUI but WinSAT still runs and stores results in XML format (%windir%\Performance\WinSAT\DataStore).

Q25
5 What is Windows Admin Center (WAC) diagnostics?

Answer : Windows Admin Center provides built-in diagnostic capabilities: Overview dashboard (CPU, memory, disk, network in real-time and historical), Event viewer (local and remote events in browser), Performance tools (counters, historical charts), Storage (SMART data, disk health, volumes), Network (interfaces, connections), Processes (running processes with resource usage), Firewall rules, Services status, Registry editor, PowerShell console, Scheduled tasks, Installed extensions/updates. WAC simplifies remote server management, replacing many MMC snap-ins. Available for Hyper-V clusters, failover clusters, and individual servers.

Q25
6

What is log analytics for Windows Server?

Answer
:

Log Analytics (Azure Monitor) collects Windows Server event logs, performance counters, and IIS logs for centralized analysis and alerting. Deploy: install Log Analytics Agent (MMA) or Azure Monitor Agent (AMA) on Windows Servers. Configure data collection: Event Logs (channels, minimum severity), Performance Counters (counters and sample intervals). Query with KQL: search all events from specific computers, create alert rules, build dashboards, correlate security events. Log Analytics Workspace stores data – configure retention period. Required for: Azure Security Center, Azure Sentinel, Update Management, Change Tracking. Microsoft recommends AMA over legacy MMA agent.

Q25
7

What is the Windows Sysinternals suite?

Answer
:

Sysinternals (acquired by Microsoft) provides 70+ advanced utilities for Windows administration and troubleshooting: Process Explorer (detailed process/thread/handle viewer – replacement for Task Manager), Process Monitor (file/registry/network/process activity), Autoruns (all auto-starting programs – find malware persistence), TCPView (network connections per process), Sigcheck (verify file signatures), PsExec (remote execution), AccessChk (permissions auditing), Handle (open file handles), RAMMap (memory usage analysis), VMMap (process virtual memory), BgInfo (desktop info display). Available at docs.microsoft.com/sysinternals. Essential toolkit for every Windows admin.

Q25
8

How do you troubleshoot Active Directory replication?

Answer
:

AD replication troubleshooting tools: repadmin /showrepl (show replication status all DCs), repadmin /replsummary (summary of replication health), repadmin /syncall /AdeP (sync all partitions), repadmin /kcc (force KCC to recalculate topology), dcdiag /test:replications (comprehensive replication test), dcdiag /test:netlogons, Event Viewer > Directory Service log. Common errors: error 8453 (access denied), 8457 (replication disabled), 1722 (RPC server unavailable – often network/DNS issue), 1396 (Logon Failure – time difference?). Check: DNS (nslookup -type=SRV _ldap._tcp.dc._msdcs.domain), firewalls (AD replication ports: 135 RPC, 49152+ dynamic, 389 LDAP, 88 Kerberos).

Q25
9

What are common Windows Server boot failure scenarios and fixes?

Answer : Boot failures: Bootloader corruption – bootrec /fixmbr, bootrec /fixboot, bootrec /rebuildbcd (run from Windows RE). System file corruption – DISM /RestoreHealth from RE, SFC /scannow offline. Bad driver causing BSOD – boot Safe Mode, disable driver in Device Manager or bcdedit /set {default} safeboot minimal. Missing or corrupt NTLDR/bootmgr – reinstall from installation media. AD DS boot issues – boot into DSRM (F8 > Directory Services Restore Mode), fix database with ntdsutil. Partition issues – diskpart in RE. Always try automatic repair first: boot RE > Troubleshoot > Advanced Options > Startup Repair.

Q26₀ How do you use ProcMon and Process Explorer together?

Answer : Combined investigation: Process Explorer identifies which processes are consuming excessive resources (CPU, handles, memory) or behaving unexpectedly. Process Explorer shows: parent-child process relationships, security context (SYSTEM, user), loaded DLLs, process strings, TCP/IP connections per process. Once you identify the process with Process Explorer, use ProcMon to trace its detailed activity: filter by PID to see exactly which files, registry keys, and network connections it accesses. This combination is ideal for: malware analysis (identify suspicious behaviors), application troubleshooting (access denied errors, missing dependencies), and performance root cause analysis.

Q26₁ What are Windows Server diagnostic commands for AD?

Answer : AD diagnostic commands: dcdiag (comprehensive DC health tests – all tests: dcdiag /e for all DCs in forest), nltest /dsgetdc:domain (find domain controller), nltest /sc_verify:domain (verify secure channel), netdom query fsmo (identify FSMO holders), ldp.exe (GUI LDAP query tool – test connectivity), klist (Kerberos ticket cache – klist tickets, klist purge), netlogon.log (enable: nltest /dbflag:0x2080ffff – diagnose logon failures), Replication Metadata: repadmin /showobjmeta (object metadata), Event ID 1004 (NETLOGON – domain joined but cannot find DC), Event ID 5 (NETLOGON – Kerberos issues).

Section 10: Windows Server Hybrid & Cloud Integration

Questions 271 – 300

Q27
1 What is Windows Server hybrid cloud integration?

Answer : Hybrid cloud integration extends on-premises Windows Server management to Azure. Key services: Azure Arc (manage on-prem servers through Azure), Azure Monitor (collect metrics and logs from on-prem servers), Azure Backup (back up on-prem workloads to Azure), Azure Site Recovery (replicate on-prem to Azure for DR), Azure AD Connect (sync AD identities to Azure AD), Windows Admin Center with Azure integration, Azure File Sync (extend on-prem file servers to Azure), Update Management (patch on-prem servers from Azure Automation), and Azure Defender for Servers (extended threat detection for on-prem).

Q27
2 What is Azure Arc for servers?

Answer : Azure Arc enables managing non-Azure servers (on-premises, other clouds) through the Azure control plane. Install the Connected Machine agent on Windows/Linux servers to: enroll them in Azure, apply Azure Policy, enable Azure Monitor and Defender, manage with Azure Automation Update Management, use Azure RBAC for server access, assign Azure tags for cost management, and use Azure Automanage for configuration best practices. Servers appear in Azure Portal as Arc-enabled servers. No VPN required for outbound HTTPS (443) connectivity. Free base service; pay for Azure services used.

Q27
3 What is Azure Backup for Windows Server?

Answer : Azure Backup protects Windows Server workloads to Azure. MARS Agent (Microsoft Azure Recovery Services Agent): backs up files/folders and system state from any Windows Server to Recovery Services Vault. Azure Backup Server (MABS): backs up VMs, SQL Server, SharePoint, Exchange to vault. Azure Backup for VMs: agent-less backup of Azure VMs and Arc-enabled VMs on-premises. Recovery options: item-level recovery (files), volume recovery, system state restore, bare-metal recovery. Backup policies: retention up to 99 years, multiple backups per day, GRS (geo-redundant) vault replication.

Q27
4 What is Azure File Sync?

Answer : Azure File Sync transforms Windows Server into a cache for Azure file shares. Key features: cloud tiering (infrequently accessed files moved to Azure, kept as sparse reparse points on server — server disk freed), multi-site sync (multiple servers sync with same Azure file share — branch office collaboration), central backup (back up file share in Azure), DR (restore server from Azure file share if server fails). Deploy: install AFS agent on Windows Server, create Storage Sync Service in Azure, create sync group, add server endpoints. Requires: Azure Files share (Standard or Premium tier).

**Q27
5** What is Azure Automation Update Management?

Answer : Azure Automation Update Management provides unified update management for Windows and Linux servers across on-premises, Azure, and other clouds. Architecture: servers with Log Analytics agent send update data to workspace □ Automation Account evaluates compliance □ deployment schedules install updates using Automation Runbooks + Hybrid Runbook Workers. Features: update compliance reporting, pre/post maintenance scripts, maintenance windows, reboot controls, targeting by tag/computer group. Benefits: single console for all servers, scheduled patching, compliance reporting for auditors. No WSUS server required for cloud-native patch management.

**Q27
6** What is Windows Admin Center Azure integration?

Answer : Windows Admin Center provides Azure hybrid services integration: Azure Backup (configure directly from WAC), Azure Site Recovery (set up replication from WAC), Azure Monitor (enable monitoring and configure alerts), Azure Defender for Servers (enroll server in Defender), Azure File Sync (deploy and manage from WAC), Azure Network Adapter (create point-to-site VPN), Azure Extended Network (extend on-prem subnet to Azure), Update Management (configure Azure Update Management), Azure Arc (register server with Arc), and Cluster creation wizard (create HCI cluster with Azure Stack HCI registration).

**Q27
7** What is Azure Active Directory Application Proxy?

Answer : Azure AD Application Proxy enables secure remote access to on-premises web applications without VPN — users access apps through Azure AD authentication. Components: Application Proxy Connector (lightweight agent installed on Windows Server on-prem, outbound HTTPS 443 connection to Azure), Azure AD application registration (configures the app URL and access policies). Features: single sign-on (SSO) to on-prem apps using Azure AD credentials, Conditional Access integration (require MFA, compliant device), pre-authentication before reaching on-prem app, and no inbound firewall rules needed (connector initiates outbound connection). Replaces WAP/AD FS for many scenarios.

**Q27
8** What is Microsoft Endpoint Manager (Intune) for Windows Server?

Answer : While Intune primarily manages client devices (Windows 10/11, iOS, Android, macOS), it integrates with Windows Server: co-management (SCCM + Intune for Windows 10/11 clients), Intune manages Windows Server 2019+ enrolled via Azure Arc (policies, compliance), and Windows Admin Center with Intune connection for enhanced management. For pure server management, Azure Arc + Azure Policy + Azure Automation is the primary path. Intune compliance policies can require Azure Arc-enabled servers to meet specific configurations before allowing Azure AD Conditional Access.

Q27
9 What is Azure Monitor for on-premises Windows Server?

Answer : Azure Monitor collects metrics and logs from on-premises Windows Servers via agents: Log Analytics Agent (MMA – legacy), Azure Monitor Agent (AMA – modern, recommended). Collected data: Windows Event Logs (System, Application, Security), Performance Counters (CPU, memory, disk, network), IIS Logs, Custom logs, and Syslog. Features: Log Analytics queries (KQL), metric charts and dashboards, alerts (on log queries or metrics), workbooks (rich reporting), VM Insights (performance and dependency maps), and integration with Defender for Cloud and Sentinel. Data stored in Log Analytics Workspace – configure retention.

Q28
0 What is Azure Security Center / Defender for Cloud on-premises?

Answer : Microsoft Defender for Cloud extends to on-premises servers via Azure Arc. Features for Arc-enrolled servers: Secure Score (security posture), Security recommendations (missing patches, insecure configurations), Regulatory compliance (CIS, PCI DSS, NIST), Defender for Servers plan (EDR via MDE integration, vulnerability assessment, threat detection, behavioral analytics, Just-in-Time VM Access). Defender for Cloud is free for basic features; Defender for Servers P1/P2 plans required for advanced threat protection. Onboard: Arc enrollment automatically enables Defender for Cloud baseline features.

Q28
1 What is Windows Server on Azure (Azure VMs)?

Answer : Azure Virtual Machines support Windows Server 2019, 2022, and other versions as first-class citizens. Azure-specific features for Windows Server VMs: Accelerated Networking (SR-IOV), Ultra Disk support, Azure Dedicated Host, Azure Spot VMs, Custom Script Extension (run PowerShell scripts post-deployment), Azure VM Image Builder (create custom images), Automatic Guest Patching (Azure handles Windows updates), Automanage (automatically applies Azure best practices), Trusted Launch (vTPM, Secure Boot, Boot Integrity monitoring), and Automatic Extension upgrade. License: Azure Hybrid Benefit saves 40-80% using existing licenses.

Q28
2 What is Windows Server Subscription (Azure Hybrid Benefit)?

Answer : Azure Hybrid Benefit allows using on-premises Windows Server licenses with active Software Assurance (SA) on Azure VMs – paying only for the base compute rate without Windows Server license cost. Without Hybrid Benefit: pay compute + Windows Server license. With Hybrid Benefit: pay compute only. Savings: 40-80% for Windows Server VMs. Also applies to: SQL Server on Azure VMs, Azure SQL Database, Azure SQL Managed Instance (up to 85% savings). Activate per VM in Azure Portal or ARM template. Eligible licenses: Windows Server Standard (2 core licenses for 1 VM), Datacenter (unlimited VMs).

Q28 3 What is the difference between Azure VM and on-premises Windows Server management?

Answer : Key differences: Patching (Azure: Automatic Guest Patching available, or Azure Update Management; On-prem: WSUS, SCCM, or Azure Update Management via Arc). Identity (Azure: Azure AD + AD DS in VMs; On-prem: AD DS). Backup (Azure: Azure Backup built-in; On-prem: Azure Backup agent or MABS). Monitoring (Azure: Azure Monitor built-in metrics; On-prem: requires agent). Networking (Azure: NSGs, Azure Firewall, no on-prem subnets; On-prem: traditional switching/routing). Storage (Azure: managed disks, no physical management; On-prem: physical disk management). Scale (Azure: resize in minutes; On-prem: hardware procurement).

Q28 4 What is Azure Stack HCI registration?

Answer : Azure Stack HCI clusters must be registered with Azure within 30 days of creation. Registration enables: Azure Hybrid Benefit pricing, Azure billing, access to Azure cloud services (Update Management, Backup, Monitoring, Defender), and full feature access. Register: Register-AzStackHCI PowerShell cmdlet (requires Azure subscription and Azure Arc resource provider). After registration: cluster appears as Azure Arc resource, Azure portal management available. Annual cloud service subscription: \$10/physical core/month. If registration lapses: Hyper-V continues running but new cluster features and support are unavailable.

Q28 5 What is Windows Server 2022 Azure Edition?

Answer : Windows Server 2022 Datacenter Azure Edition is optimized specifically for Azure and Azure Stack HCI. Exclusive features: Hotpatch (apply security updates without rebooting – reduces reboots up to 60%), SMB over QUIC (file shares accessible over internet without VPN – HTTPS-like security for SMB), Extended Network (stretch on-prem subnets to Azure), and future Azure-exclusive features first. Hotpatch details: uses Windows Update cumulative update baselines every 3 months (requires reboot), hotpatches applied monthly between baselines (no reboot). Available as Azure VM image and on Azure Stack HCI.

Q28 6 What is Windows Server 2022 security improvements?

Answer : Windows Server 2022 security enhancements: Secured-core Server (hardware-backed security: TPM 2.0, Secure Boot, DRTM, VBS, HVCI — recommended for Tier 0), TLS 1.3 enabled by default (removes weak cipher suites), DNS over HTTPS (DoH) support (encrypted DNS queries), SMB AES-256 encryption (stronger than AES-128 in previous versions), SMB compression (reduces network bandwidth for file transfers), Credential Guard on by default for supported hardware, HTTPS for Windows Remote Management, and improved BitLocker support. Secured-core Server designation requires specific OEM hardware meeting security requirements.

Q28
7 What is Azure Automanage for Windows Server?

Answer : Azure Automanage automatically applies Azure management best practices to Windows Server VMs (Azure or Arc-enabled on-prem). Automatically enables: Azure Backup, Azure Monitor (VM Insights), Defender for Cloud, Change Tracking, Update Management, Azure Guest Configuration (policy compliance), and diagnostic settings. Best Practices profile: Dev/Test (less frequent backup, smaller retention) or Production (more robust protection). Benefits: one-click onboarding, automatic drift correction (if service is accidentally disabled, Automanage re-enables it), compliance reporting. Reduces management overhead for standard Windows Server workloads.

Q28
8 What is Windows Admin Center in Azure?

Answer : Windows Admin Center is available as an integrated tool within the Azure Portal for managing Azure Windows Server VMs — no need to install WAC separately. Access: Azure VM > Operations > Windows Admin Center. Features: manage Windows Server directly from Azure Portal (no public IP, no RDP needed), file system access, PowerShell console, Event Viewer, Services, Storage, Network, Device Management. Communication uses Azure Relay (no inbound firewall rules). Works for VMs in VNets without public endpoints — connectivity via Azure control plane. Preview feature — available for Windows Server 2016+ Azure VMs.

Q28
9 What is Azure Image Builder for Windows Server?

Answer : Azure Image Builder (AIB) automates the creation of custom Windows Server VM images. Process: start from a base image (Azure Marketplace Windows Server), apply customizations (install software, configure settings, run scripts), capture the final image to Shared Image Gallery or managed image. Benefits: automated, repeatable image creation (no more manual golden image maintenance), consistent baseline for all VMs, version management, geo-replication of images. Uses Packer under the hood. Define in JSON template or Bicep. Integrate with Azure DevOps for CI/CD image pipelines.

Q29
0 What is Change Tracking and Inventory for Windows Server?

Answer : Azure Change Tracking (part of Azure Automation) monitors changes to Windows Server configurations: software installations/removals, Windows services (start/stop, configuration changes), Registry keys (monitor specific keys), Files (track modifications to critical files), and Linux daemons. Inventory feature: provides current state of all installed software, Windows features, services, registry keys, and files across all enrolled servers. Data stored in Log Analytics Workspace. Alert on: unauthorized software installation, service disablement, critical file changes. Essential for: security compliance, change management, troubleshooting (identify what changed before an issue occurred).

**Q29
1** What is Azure Policy for Windows Server governance?

Answer : Azure Policy enforces governance rules on Azure VMs and Arc-enabled on-premises servers. Built-in Windows Server policies: 'Audit Windows machines missing security update', 'Windows machines should have Windows Defender enabled', 'Audit Windows Server 2019 or later', 'Configure time zone', 'Windows Firewall should be enabled'. Guest Configuration (now Azure Machine Configuration): deploy policies that check server configurations (local users/groups, security settings, file permissions) and remediate drift. Assign policies at subscription, resource group, or management group scope. Non-compliant servers show in Policy compliance dashboard.

**Q29
2** What is the Windows Server Hybrid Deployment Checklist?

Answer : Key checklist for hybrid Windows Server deployments: 1) Deploy Azure AD Connect for hybrid identity (PHS/PTA/Federation). 2) Configure Azure AD Application Proxy for on-prem app access. 3) Install Log Analytics Agent for monitoring. 4) Configure Azure Backup for data protection. 5) Set up Azure Site Recovery for DR. 6) Register servers with Azure Arc. 7) Enable Defender for Cloud and Defender for Servers. 8) Configure Update Management for unified patching. 9) Deploy Azure File Sync for file server cloud tiering. 10) Enable Change Tracking and Inventory. 11) Apply Azure Policy for governance. 12) Configure WAC with Azure integrations.

**Q29
3** What is Microsoft's future direction for Windows Server?

Answer : Microsoft's roadmap: Continued investment in Windows Server 2022 with cumulative feature updates via Windows Update (not annual releases). Azure Edition innovations: Hotpatch, SMB over QUIC, and future cloud-optimized features first delivered to Azure Edition. Azure Stack HCI as the primary on-premises platform (annual subscription, Azure-integrated, continuously updated). Azure Arc becoming the primary management plane for all servers (on-prem, multi-cloud, edge). Gradual shift to Azure AD (Entra ID) with Azure AD DS replacing some traditional AD DS scenarios. .NET evolution: cross-platform .NET 8/9+ gradually replacing Windows-only .NET Framework. Containers: nanoserver and Windows Server Core as container base images for cloud-native workloads.